FACULTY OF ENGINEERING AND ARCHITECTURE
DEPARTMENT OF COMPUTER ENGINEERING
COMPUTER ENGINEERING

HR Management System Requirements Specification

Version 1.0

Prepared by:

Albi Idrizi
Andja Cane
Elvis Pisli
Jonida Shehi
Melisa Laçi
Sindi Pepa
Luis Kapaj

Received by:

M.Sc Shkumbin Fida
M.Sc Megi Nako

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1. Executive Summary

1.1 Project Overview

The HR Management System (HRMS) is a comprehensive web-based solution designed to modernize and streamline the management of human resource operations in small and medium-sized businesses. Many organizations still rely on spreadsheets, paper forms, and disconnected systems to manage employee information, leave requests, attendance, payroll, and performance reviews. These traditional methods are inefficient, time consuming, prone to human error, and often create delays in administrative processes.

The proposed HRMS addresses these challenges by providing a centralized digital platform that automates and integrates key HR activities such as employee data management, leave and overtime approval workflows, payroll processing, performance evaluations, onboarding, task management, and internal communication. The system is accessible through a web browser and supports different user roles including employees, team leads, department managers, HR managers, and system administrators. By digitizing and automating HR operations, the HRMS improves organizational efficiency, enhances employee self-service, supports better decision-making, and strengthens compliance and record accuracy.

1.2 Purpose and Scope of this Specification

The purpose of this project is to develop a web application that facilitates and automates the management of human resources within an organization. The system is intended to replace manual, paper-based, and fragmented HR processes with a secure, integrated, and user-friendly platform. It will enable employees to manage their own HR related activities, such as leave requests and payslip access, while allowing managers and HR staff to efficiently oversee approvals, payroll, reporting, employee performance, and compliance related activities.

This Software Requirements Specification (SRS) defines the overall scope, context, user roles, assumptions, constraints, and dependencies of the HR Management System. It serves as a reference for developers, stakeholders, instructors, and users by clearly documenting the intended functionality and operational environment of the software.

2. Product/Service Description

This section describes the general factors that influence the HR Management System and its requirements. It provides the background and context needed to understand why the system is being developed and what organizational needs it addresses.

2.1 Product Context

The HR Management System is designed for small and medium sized businesses (SMBs) and serves as a standalone web-based system accessible to employees, team leads, department

managers, HR managers, and system administrators. Its purpose is to centralize and automate the core functions of human resource management, including employee lifecycle management, attendance and leave tracking, payroll administration, performance reviews, onboarding, document management, and internal communication.
The system enhances collaboration between employees, supervisors, HR personnel, and management by ensuring that all HR-related data and workflows are handled in one unified platform. It supports role-based access control so that each type of user can perform only the tasks relevant to their responsibilities. By replacing fragmented and manual processes, the HRMS improves data consistency, operational transparency, and decision-making across the organization.

2.2 User Characteristics

1. Employee

- View and manage personal profile information.
- Submit leave requests and overtime requests.
- Clock in and clock out for attendance tracking.
- View salary history and download payslips.
- Access company announcements and notifications.
- Upload and manage personal documents.
- Book desks and complete onboarding tasks.
- View assigned tasks and track their progress.

2. Team Lead

- Perform all Employee level functions.
- View time logs and leave requests of team members.
- Recommend or review leave and overtime requests for first-level approval.
- Assign tasks and monitor team performance.
- Access team-related analytics and summaries.

3. Department Manager

- Perform all Team Lead level functions.
- Oversee all employees within a department.
- Review and recommend leave or overtime requests for department members.
- Access department level reports and workforce summaries.
- Publish or manage department-specific announcements.

4. HR Manager

- Perform all Department Manager level functions.
- Manage employee records, salary records, and payroll processing.
- Provide final approval for leave and overtime requests.
- Conduct and manage performance reviews.

- Configure leave types, holiday calendars, onboarding templates, and document categories.
- Generate reports related to attendance, payroll, headcount, and employee performance.

5. SuperAdmin

- Perform all HR Manager level functions.
- Manage user accounts, authentication, and role assignments.
- Access and review system wide audit logs.
- Configure global system settings.
- Perform administrative actions such as password resets and infrastructure-level

These users are expected to have basic digital literacy and the ability to interact with a modern web application. However, the complexity of system usage varies depending on the user role, with administrative roles requiring more advanced interaction with reports, configurations, and approval workflows.

2.3 Assumptions

- Employees, team leads, managers, and HR personnel will receive sufficient training to use the application effectively.
- The organization has the necessary IT infrastructure, including computers, internet access, and secure devices, to support system use.
- Employee records, salary information, leave balances, and attendance data will be entered accurately and maintained regularly.
- Employees will actively use the system for self service functions such as leave requests, document access, and attendance logging.
- Team leads and department managers will review and process approvals in a timely and responsible manner.
- HR managers will maintain updated employee records, payroll configurations, and organizational settings.
- SuperAdmin users will ensure proper user-role assignment and system-level governance.
- The organization will adopt the system as the primary HR platform instead of continuing parallel paper-based workflows.

2.4 Constraints

- The HR Management System requires a stable internet connection because it is a web-based platform.
- The effectiveness of the system depends heavily on user adoption and staff training.

- The performance and reliability of the application depend on the organization's hardware, network stability, and browser compatibility.
- Payroll accuracy depends on correct salary data, leave balances, overtime records, and deductions being entered into the system.
- Approval workflows are dependent on the timely response of team leads, department managers, and HR managers.
- Features such as PDF payslip generation, notifications, and document management rely on proper configuration and implementation of supporting modules.
- The system is constrained by the limitations of the selected web technologies, frameworks, and available development time.
- Security and privacy controls must be enforced to protect employee and payroll data from unauthorized access.
- The first version of the system is limited to a web-based platform only, without dedicated mobile applications.

2.5 Dependencies

- Employee profile management depends on the successful creation and maintenance of employee records by HR or administrative staff.
- Leave requests and overtime approvals depend on the active participation of team leads, department managers, and HR managers.
- Payroll processing depends on the availability and correctness of salary records, attendance logs, approved leave, approved overtime, and deductions.
- Time tracking depends on employees consistently clocking in and clocking out through the system.
- Performance review workflows depend on managers and employees completing review forms and assessments on time.
- Onboarding processes depend on HR managers creating templates and assigning checklist items to new employees.
- Task management depends on team leads and managers creating and updating tasks for employees.
- Document management depends on users uploading required documents and HR maintaining document categories and validity dates.
- Notification and announcement features depend on administrators and HR staff publishing relevant updates through the platform.
- Audit logging and compliance tracking depend on all system interactions being properly recorded and preserved by the backend system.

3. Requirements

3.1 Functional Requirements

1. Employee Management

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-EMP-001	Create Employee Record	The system shall allow authorized users (HR Manager, SuperAdmin) to create a new employee record with first name, last name, email, phone, address, date of birth, national ID, hire date, job title, department, team, manager, employment type, and standard work hours.	1	31/03/2026	Sindi Pepa/ Jonida Shehi
FR-EMP-002	View Employee Directory	The system shall provide a paginated, searchable employee directory accessible to all authenticated users. The directory shall display name, job title, department, team, status, and profile photo.	1	31/03/2026	Jonida Shehi/ Andja Cane
FR-EMP-003	Update Employee Profile	The system shall allow authorized users to update employee profile fields. Optimistic concurrency control (RowVersion) shall prevent simultaneous conflicting updates.	1	31/03/2026	Sindi Pepa/ Melisa Laci
FR-EMP-004	Soft Delete Employee	The system shall support soft deletion of employee records, marking them as deleted with a timestamp and the deleting user's identity, without physically removing data from the database.	1	31/03/2026	Andja Cane/ Luis Kapaj
FR-EMP-005	Transfer Employee Between Departments	The system shall allow HR Manager or SuperAdmin to transfer an employee to a different department, recording the transfer in the DepartmentTransferHistory table.	2	31/03/2026	Melisa Laci/ Elvis Pisli

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-EMP-006	View Employee Detail Page	The system shall provide a detailed employee profile page showing personal information, employment details, salary history, documents, and performance data, respecting role-based visibility rules.	1	31/03/2026	Albi Idrizi/ Elvis Pisli

2. Leave Management

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-LV-001	Configure Leave Types	The system shall allow HR Managers to create and update leave types with name, default days per year, paid/unpaid flag, and carry-over policy.	1	31/03/2026	Luis Kapaj/ Elvis Pisli
FR-LV-002	Initialize Leave Balances	The system shall allow HR Managers to initialize annual leave balances for all active employees based on configured leave-type defaults.	1	31/03/2026	Andja Cane/ Sindi Pepa
FR-LV-003	Submit Leave Request	The system shall allow employees to submit a leave request specifying leave type, start date, end date, reason, and optional file attachment.	1	31/03/2026	Jonida Shehi/ Luis Kapaj
FR-LV-004	Two-Tier Leave Approval	The system shall implement a two-tier approval workflow: (1) Team Lead or Department Manager recommends the request; (2) HR Manager or SuperAdmin provides final approval or rejection.	1	31/03/2026	Melisa Laci/ Jonida Shehi
FR-LV-005	Reject Leave Request	The system shall allow authorized approvers to reject a leave request at any approval tier, providing mandatory comments explaining the rejection.	1	31/03/2026	Melisa Laci/ Andja Cane
FR-LV-006	View Leave Balance	The system shall allow employees to view their current leave balances by leave type for the current year, including total entitlement, used days, and remaining days.	1	31/03/2026	Elvis Pisli/ Albi Idrizi

3. Time Tracking and Overtime

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-TT-001	Clock In	The system shall allow employees to record their clock-in time for the current day. Only one open (un-clocked-out) session shall be permitted at a time.	1	31/03/2026	Sindi Pepa/ Elvis Pisli
FR-TT-002	Clock Out	The system shall allow employees to record their clock-out time, automatically calculating the duration in minutes from the start time.	1	31/03/2026	Elvis Pisli/ Luis Kapaj
FR-TT-003	Daily and Weekly Summaries	The system shall provide daily summaries (total hours per day) and weekly summaries (total hours, days worked, average daily hours) for each employee.	1	31/03/2026	Jonida Shehi/ Albi Idrizi
FR-TT-004	Time-Log Modification Request	The system shall allow employees to submit modification requests for existing time logs (to correct errors), subject to approval by their manager.	2	31/03/2026	Sindi Pepa/ Andja Cane
FR-TT-005	Team Time-Log View	The system shall provide Team Leads and above with a view of their team members' time logs, including clock-in/out times, durations, and daily totals.	1	31/03/2026	Andja Cane/ Elvis Pisli
FR-OT-006	Submit Overtime Request	The system shall allow employees to submit manual overtime requests specifying date, duration, overtime type (Weekday, Weekend, Holiday), and reason.	1	31/03/2026	Melisa Laci/ Luis Kapaj
FR-OT-007	Automatic Overtime Detection	The system shall automatically detect overtime when an employee's clock-out time results in daily hours exceeding their configured standard work hours, creating an overtime record linked to the time log.	2	31/03/2026	Albi Idrizi/ Luis Kapaj

4. Salary and Payroll

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-SAL-001	Manage Salary Records	The system shall allow HR Managers to create and view salary records for employees, specifying base salary, currency, effective date, and notes. Optimistic concurrency control shall be enforced.	1	31/03/2026	Jonida Shehi/ Sindi Pepa
FR-SAL-002	Manage Allowances	The system shall support adding recurring allowances (Housing, Transport, Meal, Other) to an employee's salary, each with a name, amount, and type.	1	31/03/2026	Andja Cane/ Jonida Shehi
FR-SAL-003	Manage Deductions	The system shall support deductions with a review workflow. Deductions can be proposed and must be reviewed before becoming active.	1	31/03/2026	Luis Kapaj/ Jonida Shehi
FR-SAL-004	Manage Bonuses	The system shall allow HR Managers to add one-time or recurring bonuses to employee records with amount, reason, and effective date.	2	31/03/2026	Luis Kapaj/ Sindi Pepa
FR-PAY-001	Create Payroll Run	The system shall allow HR Managers to create a payroll run for a specific month and year. The run is initialized in Draft status.	1	31/03/2026	Albi Idrizi/ Andja Cane
FR-PAY-002	Generate Payslips	The system shall generate individual payslips for all active employees in a payroll run, calculating base salary, allowances, bonuses, deductions, paid/unpaid leave adjustments, gross pay, and net pay.	1	31/03/2026	Albi Idrizi/ Melisa Laci
FR-PAY-003	Review and Edit Payslips	The system shall allow HR Managers to review individual payslips and make manual adjustments before finalization. Concurrency control prevents conflicting edits.	1	31/03/2026	Luis Kapaj/ Albi Idrizi
FR-PAY-004	Finalize Payroll Run	The system shall allow authorized users to finalize a payroll run, changing its status to Finalized and recording the finalizer and timestamp. Finalized runs cannot be modified.	1	31/03/2026	Luis Kapaj/ Andja Cane

5. Performance Reviews

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-PR-001	Create Review Cycle	The system shall allow HR Managers to create named review cycles with start and end dates, review type (Annual, Mid-Year, Quarterly, Probation), and status.	1	31/03/2026	Andja Cane/ Melisa Laci
FR-PR-002	Self-Assessment	The system shall allow employees to complete a self-assessment within an active review cycle, providing self-assessment text and a numeric self-rating.	1	31/03/2026	Jonida Shehi/ Elvis Pisli
FR-PR-003	Manager Review	The system shall allow managers to provide a manager review with narrative text, numeric rating, and assessment of employee goals.	1	31/03/2026	Sindi Pepa/ Albi Idrizi
FR-PR-004	Goal Tracking	The system shall support the creation and tracking of performance goals within each review, with title, description, weight percentage, and achievement level.	2	31/03/2026	Sindi Pepa/ Elvis Pisli
FR-PR-005	Overall Rating Calculation	The system shall compute an overall rating based on self-assessment, manager rating, and goal achievement, with configurable weighting.	2	31/03/2026	Sindi Pepa/ Jonida Shehi

6. Task Management

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-TK-001	Create Task	The system shall allow Team Leads and above to create tasks with title, description, assignee, priority (Low, Medium, High, Urgent), due date, category, and tags.	1	31/03/2026	Jonida Shehi/ Andja Cane
FR-TK-002	Update Task Status	The system shall allow task assignees and creators to update task status through the workflow: ToDo, InProgress, InReview, Done, Cancelled.	1	31/03/2026	Sindi Pepa/ Melisa Laci
FR-TK-003	Task Comments	The system shall support threaded comments on tasks, allowing assignees and stakeholders to communicate about task progress.	2	31/03/2026	Andja Cane/ Luis Kapaj

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-TK-004	Filter and Search Tasks	The system shall provide filtering by status, priority, assignee, category, and due-date range, with full-text search on title and description.	1	31/03/2026	Melisa Laci/ Elvis Pisli

7. Onboarding

Req#	Requirements	Comments	Priorit	Data Rvwd	SME Reviewed / Approved
FR-OB-001	Create Onboarding Template	The system shall allow HR Managers to create onboarding templates with a name, description, and ordered list of checklist items (title, description, assignment).	1	31/03/2026	Luis Kapaj/ Elvis Pisli
FR-OB-002	Complete Checklist Items	The system shall allow designated completers (employee, manager, or HR) to mark individual checklist items as completed.	1	31/03/2026	Andja Cane/ Sindi Pepa
FR-OB-003	Assign Onboarding Checklist	The system shall allow HR Managers to assign an onboarding template to a new employee, creating an OnboardingChecklist with individual checklist items.	1	31/03/2026	Melisa Laci/ Andja Cane
FR-OB-004	Onboarding Progress Dashboard	The system shall provide a dashboard view showing onboarding progress for all active checklists, with completion percentages and overdue items.	2	31/03/2026	Elvis Pisli/ Albi Idrizi

8. Document Management

Req#	Requirement	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-DOC-001	Upload Document	The system shall allow authorized users to upload documents associated with an employee record, specifying category, notes, and optional expiry date. File metadata (name, size, content type) shall be recorded.	1	31/03/2026	Jonida Shehi/ Albi Idrizi

Req#	Requirement	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-DOC-002	Download Document	The system shall allow authorized users to download employee documents. Access is restricted based on role (employees can download their own; HR can download any).	1	31/03/2026	Sindi Pepa/ Andja Cane
FR-DOC-003	Document Categories	The system shall support configurable document categories (e.g., Contract, ID, Certificate, Policy) managed by HR.	1	31/03/2026	Andja Cane/ Elvis Pisli
FR-OB-004	Onboarding Progress Dashboard	The system shall provide a dashboard view showing onboarding progress for all active checklists, with completion percentages and overdue items.	2	31/03/2026	Melisa Laci/ Luis Kapaj
FR-DOC-005	Document Expiry Tracking	The system shall track document expiry dates and surface upcoming expirations in the HR dashboard.	3	31/03/2026	Albi Idrizi/ Luis Kapaj

9. Announcements

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-AN-001	Create Announcement	The system shall allow HR Managers and above to create announcements with title, body, priority (Low, Normal, High, Urgent), target department (optional for company-wide), pin status, and publish date.	1	31/03/2026	Sindi Pepa/ Elvis Pisli
FR-AN-002	View Announcements	The system shall display announcements to employees in reverse-chronological order, with pinned announcements at the top. Employees see company-wide announcements and those targeted to their department.	1	31/03/2026	Jonida Shehi/ Sindi Pepa
FR-AN-003	Read Receipts	The system shall track which employees have read each announcement via read receipts, enabling HR to monitor communication reach.	2	31/03/2026	Andja Cane/ Jonida Shehi
FR-AN-004	Soft Delete Announcement	The system shall support soft deletion of announcements, removing them from active views while preserving them for audit purposes.	1	31/03/2026	Luis Kapaj/ Jonida Shehi

10. Desk Booking

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-DB-001	Manage Office Hierarchy	The system shall support a three-level hierarchy: Offices contain Floors, and Floors contain Desks. Each entity has a name, status, and optional metadata.	1	31/03/2026	Andja Cane/ Melisa Laci
FR-DB-002	Book a Desk	The system shall allow employees to book an available desk for a specific date. The system shall prevent double-booking (two employees booking the same desk on the same date).	1	31/03/2026	Jonida Shehi/ Elvis Pisli
FR-DB-003	Cancel Booking	The system shall allow employees to cancel their own desk bookings. HR and SuperAdmin can cancel any booking.	1	31/03/2026	Sindi Pepa/ Albi Idrizi
FR-DB-004	View Desk Availability	The system shall provide a view of desk availability for a given date and floor, showing which desks are available, booked, or out of service.	2	31/03/2026	Melisa Laci/ Albi Idrizi

11. Notifications

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-NT-001	In-App Notifications	The system shall generate in-app notifications for significant events: leave request status changes, task assignments, overtime approvals, announcement publications, onboarding assignments, and payroll finalization.	1	31/03/2026	Albi Idrizi/ Elvis Pisli
FR-NT-002	Mark as Read	The system shall allow users to mark individual notifications as read or mark all notifications as read in bulk.	1	31/03/2026	Luis Kapaj/ Elvis Pisli
FR-NT-003	Email Notification Preferences	The system shall allow users to configure which notification types of trigger email delivery in addition to in-app notifications.	3	31/03/2026	Elvis Pisli/ Albi Idrizi

12. Self-Service Document Generation

Req#	Requirements	Comments	Priority	Data Rvwd	SME Reviewed / Approved
FR-SS-001	Generate Employment Verification Letter	The system shall allow employees to generate a formal employment verification letter containing their name, position, department, hire date, and employment status.	2	31/03/2026	Melisa Laci/ Sindi Pepa
FR-SS-002	Generate Salary Certificate	The system shall allow employees to generate a salary certificate containing their current base salary, allowances, and total compensation.	2	31/03/2026	Andja Cane/ Jonida Shehi

3.2 Non-Functional Requirements

3.2.1 Product Requirements

a) User Interface Requirements

The user interface for the HR Management System (HRMS) shall be web-based and compatible with all modern browsers, allowing access from both desktop and mobile devices. The interface is divided into several main modules based on user roles.

1. Dashboard Interface (Admin / HR Manager):

a) A top navigation bar containing:

- Current date and time
- Logged-in user role (e.g., Super Admin)
- Notifications icon
- User profile menu

b) A sidebar navigation menu containing sections such as:

- Dashboard
- Employees
- Departments
- Teams
- Time Tracking
- Leave
- Overtime
- Salary
- Documents
- Announcements
- Reports
- Audit Log
- Settings

*****The main dashboard body will include:

a) Summary cards displaying key metrics such as:

- Total Users
- Active Users
- Inactive Users

- Roles in Use

b) Visual charts such as:

- Users by Role (graph representation)
- User Activity Summary (active vs inactive users)
- Additional statistics cards:
  - Total Employees
  - New Hires
  - Pending Leave Requests
  - Pending Overtime
  - Overdue Onboarding

c) Action buttons for quick operations:

- Add Employee
- Create Announcement
- Run Report

2. Employee Interface:

a) A simplified sidebar containing:

- Dashboard
- My Profile
- Attendance / Time Tracking
- Leave Requests
- Salary Details
- Documents

b) The main body will display:

- Personal information overview
- Attendance summary
- Leave balance and history
- Notifications and announcements

3. HR Staff Interface:

a) Sidebar options include:

- Employee Management
- Recruitment / Onboarding
- Attendance Management
- Leave Approvals
- Payroll
- Reports

b) Main interface features:

- Employee list with search and filter options
- Forms to add/edit employee details
- Leave request approvals/rejections
- Attendance tracking and updates

4.Manager Interface:

a) Sidebar options include:

- Team Overview
- Employee Performance

- Leave Approvals
- Reports

b) Main dashboard includes:
- Team performance summary
- Pending approvals (leave, overtime)
- Reports and analytics

5. Common Interface Features:

a) Responsive Design:
- The interface adapts to desktop, tablet, and mobile screens.

b) Form Validation:
- All inputs (e.g., employee data, leave requests) are validated, and error messages are shown when invalid.

c) Consistent Layout:
- Navigation structure and design elements remain consistent across all pages.

d) Interactive Elements:
- Charts, buttons, and tables provide dynamic interaction for better user experience.

e) Notifications System:
- Users receive alerts for important events (e.g., leave approval, announcements).

b) Learnability

The HRMS shall be designed to ensure that users can quickly and easily learn how to use the system with minimal training:

1. Intuitive Interface:

The system shall provide a clean and user-friendly interface with clearly labelled elements and logical layout

2. Guided Onboarding:

New users shall receive basic guidance such as tooltips, hints, or short tutorials on first use

3. Role-Based Dashboards:

Each user role shall have a customized dashboard showing only relevant features, reducing complexity

4. Consistent Navigation:

Menus, buttons, and workflows shall remain consistent across all modules

5.Minimal Training Requirement:

The system shall use familiar design patterns (forms, tables, buttons) to minimize learning effort

c) Accessibility

The HRMS shall ensure accessibility for a wide range of users:

1.Browser Compatibility:
The system shall function correctly on all major modern browsers
2. Responsive Design:
The interface shall adapt to different screen sizes (desktop, tablet, mobile)
3. Readable Design:
Proper font sizes, contrast, and spacing shall be used to enhance readability
4. Keyboard Accessibility:
Users shall be able to navigate basic functions using a keyboard
5.Text Alternatives:
Icons and images shall include labels or descriptions
6. Zoom Support:
Users shall be able to zoom in without breaking the layout

d) Efficiency

The system shall optimize performance and user productivity:
1. Fast Response Time:

- API responses ≤ 500 ms for simple operations
- ≤ 2 seconds for complex operations

2.Quick Actions:
Frequently used actions (e.g., approvals, submissions) shall be easily accessible
3. Search and Filtering:
Users shall be able to quickly search and filter employees, tasks, and records
4.Automation:
Processes such as payroll generation, notifications, and reports shall be automated
5. Optimized Data Handling:
Pagination and indexing shall be used to improve database performance

e) Dependability / Memorability

The HRMS shall ensure system reliability and ease of use:
Dependability
-The system shall ensure data integrity and prevent data loss
- All critical actions shall be logged for auditing purposes
- The system shall handle errors without crashing
- Secure authentication and authorization mechanisms shall be enforced
- Transactions shall be used for critical operations (e.g., payroll processing)
Memorability
- The interface shall maintain a consistent layout across all pages
- Standard icons and labels shall be used throughout the system
- Users shall be able to remember how to perform tasks after periods of inactivity
- Frequently used features shall be easily accessible

f) Errors

The system shall minimize user errors and provide effective recovery mechanisms:

1. **Preventive Design:**
   Input validation shall prevent incorrect or invalid data entry
2. **Clear Error Messages:**
   Error messages shall clearly explain the issue and suggest solutions
   (e.g., “Invalid email format. Please enter a valid email address”)
3. **Confirmation Dialogs:**
   Critical actions (e.g., deleting records, approving payroll) shall require user confirmation
4. **Error Logging:**
   All errors shall be logged for system monitoring and debugging
5. **Recovery Support:**
   The system shall allow users to retry actions without losing progress
6. **Graceful Handling:**
   If a feature fails, the system shall continue functioning and guide the user

g) Satisfaction

The HR Management System (HRMS) is designed to be simple, efficient, and user-friendly for all types of users, including employees, managers, HR personnel, and system administrators. The interface is intuitive and well-structured, allowing users to quickly navigate through the system and perform their tasks without requiring extensive training.

The system is organized to simplify daily HR operations such as managing employee records, submitting leave requests, tracking attendance, processing payroll, and reviewing performance. Each feature is designed to minimize the number of steps required to complete tasks, improving overall productivity.

Additionally, the system is accessible through both desktop and mobile devices, enabling users to interact with the platform anytime and from anywhere. This flexibility enhances user convenience and supports efficient workflow management.

Overall, the goal of the HRMS is to provide a smooth and hassle-free user experience that reduces manual work, improves efficiency, and increases user satisfaction across all roles within the organization.

h) Capacity

The HR Management System (HRMS) shall be designed to support the operational needs of small and medium-sized organizations. The system shall allow multiple users, including employees, managers, HR staff, and administrators, to access and use the system simultaneously.

The system shall operate in real-time, ensuring that updates such as leave approvals, task changes, or payroll processing are immediately reflected across the platform for all authorized users.

All users shall interact with a centralized database. In cases where multiple requests are made at the same time, the system shall process them efficiently using server-side handling mechanisms. Minor delays may occur under heavy load, but performance shall remain within acceptable limits.
To improve performance and reliability, the system may temporarily store user actions on the client side before synchronizing them with the server. This ensures continuity of work even during brief connectivity or processing delays.
The HRMS shall be deployed on a web server and designed to be lightweight, ensuring efficient use of system resources. Database storage requirements shall be optimized through structured data management and efficient querying techniques.

i) Security

The data stored within the HRMS database is highly sensitive, including employee personal information, salary details, and organizational records. Therefore, a high level of security must be ensured.
In compliance with data protection principles and applicable regulations, the system shall implement the following security measures:

- User passwords shall be stored in an encrypted (hashed) format, ensuring that plain-text passwords are never stored or exposed
- Secure authentication mechanisms (e.g., JWT-based authentication) shall be implemented to verify user identity
- Role-based access control (RBAC) shall restrict access to data based on user roles (Employee, Manager, HR, Admin)
- Secure communication protocols (HTTPS) shall be used to protect data during transmission
- Sensitive data shall only be accessible to authorized users
- System logs and audit trails shall track all important actions for monitoring and compliance

3.2.2 Organizational Requirements

Requirements which are a consequence of organisational policies and procedures e.g. process standards used, implementation requirements, etc

a) Availability

The HRMS shall be available 24/7 to support continuous access for employees and management.

- The system shall ensure high availability with minimal downtime
- Regular backups shall be performed to prevent data loss and ensure data recovery
- Maintenance activities shall be scheduled to minimize disruption

b) Latency

The system shall ensure fast response times for critical operations:

- Operations such as login, leave submission, approvals, and payroll processing shall have a response time of less than 2 seconds under normal conditions
- System performance shall remain stable under typical user load

c) Monitoring

The system shall include monitoring mechanisms to track performance and usage:

- A dashboard shall display system performance metrics such as uptime and user activity
- Administrators shall be able to monitor system usage and trends
- Alerts shall be generated for system failures, errors, or suspicious activities

d) Maintenance

The HRMS shall be maintained regularly to ensure reliability and security:

- Software updates shall be applied periodically
- Bug fixes and performance improvements shall be implemented
- System logs shall be maintained for troubleshooting and analysis

e) Operations

The system shall be designed for efficient day-to-day use:

- The interface shall be intuitive and easy to operate
- Users shall be able to resume usage easily after periods of inactivity
- Icons, buttons, and descriptions shall enhance usability
- Common tasks shall be simplified to reduce user effort

f) Standards Compliance

The HRMS shall comply with relevant standards and best practices:

- Employee data management shall follow HR best practices
- Payroll and reporting shall follow standard organizational procedures
- The system shall comply with data protection regulations
- Audit logs shall ensure traceability of all important actions

g) Portability

- The system shall be a web-based application accessible across different operating systems
- Users shall be able to access the system using a computer, tablet, or mobile device
- The system shall support all major web browsers
- The interface shall adapt to different screen sizes for optimal user experience

3.2.3 External Requirements

a) Protection

To protect the HRMS from unauthorized access and misuse, the system shall implement the following:

- Sensitive data such as passwords shall be encrypted using hashing techniques
- User activities shall be logged and audited for accountability
- Input validation shall prevent malicious data entry and cyberattacks
- Confirmation dialogs shall be used for critical actions
- Each user shall only access their own data or authorized data
- Role-based restrictions shall ensure privacy and security of employee information

b) Authorization and Authentication

- Users shall authenticate using email and password with secure mechanisms (e.g., JWT tokens)
- Authorization shall be role-based, ensuring access only to permitted features
- Sessions shall be managed securely to maintain user state
- Invalid login attempts shall display appropriate error messages

c) Legislative Requirements

The HRMS shall comply with applicable legal and regulatory requirements:

- Employee data shall be handled according to data protection laws
- The system shall maintain audit logs for tracking changes (e.g., salary updates, employee records)
- All critical operations shall be traceable with before-and-after values
- Reports and records shall follow organizational and legal standards

3.3 Domain Requirements

The HR Management System operates within the domain of Human Resource Management. Its primary purpose is to digitize and automate HR processes, including employee management, attendance tracking, payroll processing, performance evaluation, and internal communication.

The system shall ensure that sensitive employee data, such as personal information and salary records, is securely stored and accessible only to authorized users.

Role-based access control shall ensure that each user interacts only with features relevant to their role (Employee, Manager, HR, Admin). This improves both security and usability.

The HRMS is designed as a standalone web application and does not require integration with external systems in its initial version. However, the architecture allows for future scalability and integration if needed.

The system shall be responsive and accessible across devices, and it shall implement modern authentication and authorization mechanisms to ensure data confidentiality, integrity, and secure operations.

4. User Scenarios/Use Cases

4.1 Requirements Analysis

4.1.1 User Scenarios List

Nr	Name	Description
US_01	User logs in	Employees, Team Leads, Department Managers, HR Managers, and Super Admins log in using email and password.
US_02	User logs out	Employees, Team Leads, Department Managers, HR Managers, and Super Admins log out securely from the system.
US_03	Reset password	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can reset their password if they forget it.
US_04	Change password	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can change their password from their profile settings.
US_05	Create employee	HR Managers and Super Admins can create a new employee profile by filling in personal and employment details.
US_06	View employee directory	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can view the employee directory based on access rights.
US_07	View employee profile	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can view an employee profile.
US_08	Update employee profile	HR Managers and Super Admins can update employee information such as department, role, manager, or status.
US_09	Deactivate employee	HR Managers and Super Admins can deactivate an employee account while preserving historical records.
US_10	Transfer employee to new department	HR Managers and Super Admins can transfer an employee from one department to another.
US_11	Create department	HR Managers and Super Admins can create a new department by entering department details.
US_12	Manage department head	HR Managers and Super Admins can assign, update, or remove the department head.
US_13	Create team	HR Managers and Super Admins can create a new team within a department.
US_14	Submit leave request	Employees, Team Leads, and Department Managers can submit a leave request by selecting leave type, dates, and reason.

Nr	Name	Description
US_14	Submit leave request	Employees, Team Leads, and Department Managers can submit a leave request by selecting leave type, dates, and reason.
US_15	Recommend leave request (Tier 1)	Team Leads and Department Managers can recommend or reject leave requests from employees under their supervision.
US_16	Approve leave request (Tier 2)	HR Managers and Super Admins can approve or reject leave requests after managerial review.
US_17	Cancel leave request	Employees can cancel a submitted leave request before or after approval depending on policy.
US_18	View leave balance	Employees, Team Leads, and Department Managers can view their available leave balance.
US_19	Configure leave types	HR Managers and Super Admins can create and manage leave types and leave policies.
US_20	Clock in	Employees, Team Leads, and Department Managers can clock in to start the work session.
US_21	Clock out	Employees, Team Leads, and Department Managers can clock out to end the work session.
US_22	View time logs	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can view time logs based on role permissions.
US_23	Request time correction	Employees, Team Leads, and Department Managers can request correction for incorrect attendance records.
US_24	Approve time correction	Team Leads, Department Managers, HR Managers, and Super Admins can approve or reject time correction requests.
US_25	Submit overtime request	Employees, Team Leads, and Department Managers can submit an overtime request by entering date, duration, and reason.
US_26	Approve overtime request	Team Leads, Department Managers, HR Managers, and Super Admins can approve or reject overtime requests.
US_27	Set employee salary	HR Managers and Super Admins can create or update an employee salary record.
US_28	View salary and payslip history	Employees can view salary details and previous payslips.
US_29	Create payroll run	HR Managers and Super Admins can generate payroll for a selected month and year.
US_30	Edit payslip	HR Managers and Super Admins can modify draft payslips before payroll is finalized.
US_31	Finalize payroll	HR Managers and Super Admins can finalize payroll and lock payslips from further changes.

Nr	Name	Description
US_32	Download payslip PDF	Employees, HR Managers, and Super Admins can download finalized payslips in PDF format.
US_33	Create task	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can create tasks for manageable employees.
US_34	Update task status	Assigned Employees, Creators, and Managers can update the task status such as To Do, In Progress, or Done.
US_35	Comment on task	All users with access to the task can add comments.
US_36	Upload document	HR Managers and Super Admins can upload employee-related documents to the system.
US_37	Download document	Employees, HR Managers, and Super Admins can download documents based on access rights.
US_38	Book a desk	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can book a desk for a selected date.
US_39	Cancel desk booking	Employees, HR Managers, and Super Admins can cancel an existing desk booking.
US_40	Manage offices, floors, and desks	HR Managers and Super Admins can create, update, and manage office spaces, floors, and desks.
US_41	Create announcement	HR Managers and Super Admins can create and publish announcements for employees.
US_42	View and read announcements	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can view company announcements.
US_43	Create onboarding template	HR Managers and Super Admins can create onboarding templates for new employees.
US_44	Assign onboarding checklist	HR Managers and Super Admins can assign onboarding checklists to employees.
US_45	Complete onboarding item	Employees and assigned users can mark onboarding tasks as completed.
US_46	Create and start performance review cycle	HR Managers and Super Admins can create and launch a performance review cycle.
US_47	Submit self-assessment	Employees can submit a self-assessment during a performance review cycle.
US_48	Submit manager review	Team Leads and Department Managers can evaluate employees and submit manager reviews.
US_49	Set performance goals	Team Leads, Department Managers, and HR Managers can set employee performance goals.
US_50	Generate self-service PDF document	Employees can generate personal HR documents such as salary certificates or leave balance statements.

Nr	Name	Description
US_51	Manage holidays	HR Managers and Super Admins can create, update, and delete public holidays in the system.
US_52	View and manage notifications	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can view and manage notifications.
US_53	View calendar	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can view the organizational calendar.
US_54	View audit logs	HR Managers and Super Admins can view system audit logs and activity history.
US_55	Manage user accounts	Super Admins can create, update, activate, deactivate, and manage user accounts.
US_56	Configure settings	Employees, Team Leads, Department Managers, HR Managers, and Super Admins can configure theme and language settings.

4.1.2 User Scenarios Extended

US_01: User logs in

1. User navigates to the login page.
2. User enters email and password.
3. User clicks the Sign In button.
4. System validates the credentials against stored BCrypt hash.
5. If credentials are valid, system generates a JWT access token and refresh token.
6. System redirects the user to the role-specific dashboard.
7. System grants access based on the user role (Employee, Team Lead, Department Manager, HR Manager, Super Admin).

US_02: User logs out

1. Authenticated user clicks the profile menu in the header.
2. User selects the Logout option.
3. System revokes the active refresh token.
4. System clears tokens from browser storage.
5. System redirects the user to the login page.
6. User can no longer access protected pages unless logged in again.

US_03: Reset password

1. User navigates to the login page.
2. User clicks the Forgot Password link.
3. User enters the email associated with the account.
4. System generates a hashed reset token with 1-hour expiry.
5. System always returns a success message regardless of whether the email exists (prevents enumeration).

6. In production, system sends the reset link via email. In development mode, token is returned in the API response.
7. User opens the reset link and enters a new password (minimum 8 characters).
8. System updates the password and revokes all active refresh tokens.
9. User is redirected to the login page to sign in with the new password.

US_04: Change password

1. Authenticated user navigates to Profile > Security.
2. User enters current password and new password.
3. System verifies current password against stored hash.
4. System updates the password with new BCrypt hash.
5. System revokes all active refresh tokens.
6. User receives confirmation message.

US_05: Create employee

1. HR Manager navigates to the People section.
2. User clicks Add Employee.
3. User enters personal information (first name, last name, email, phone, date of birth, address, national ID).
4. User enters employment details (department, team, job title, employment type, hire date, password).
5. System validates all required fields and checks email uniqueness.
6. User clicks Create Employee.
7. System creates the employee record, user account (with hashed password), and initializes leave balances for all active leave types.
8. If base salary is provided, system also creates a salary record.
9. System auto-generates a unique slug from first and last name.
10. System redirects to the new employee profile page.

US_06: View employee directory

1. User navigates to the People section.
2. System retrieves employees based on role: Employee sees all active employees, Team Lead sees direct reports, Department Manager sees department, HR/Admin sees all.
3. System displays the directory in a searchable table with columns: Name, Department, Job Title, Status.
4. User can search by name or email.
5. User can filter by department, status, or employment type.
6. User selects an employee to view their full profile.

US_07: View employee profile

1. User opens the employee directory or navigates directly to /people/{slug}.
2. System retrieves employee details.
3. System displays hero card with name, avatar, status badge, role badge, email, and employment type.
4. Quick stats row shows: Department, Team, Reports To, Tenure.
5. Tabs display: Personal Info (contact, DOB, national ID), Employment (position, schedule, dates).
6. If viewer is HR/Admin, additional tabs show: Salary and Documents.

US_08: Update employee profile

1. HR Manager navigates to an employee profile.
2. User clicks the Edit button.
3. User modifies employee details such as department, job title, team, manager, role, or status.
4. System validates updated information.
5. If department is changed, system creates a transfer history record and updates manager assignment.
6. If email is changed, system syncs the linked user account email.
7. User clicks Save Changes.
8. System updates the employee record and displays confirmation.

US_09: Deactivate employee

1. HR Manager opens an employee profile.
2. User changes employee status to Terminated or Suspended.
3. System confirms the action.
4. System automatically cancels all pending/recommended leave requests with comment "Auto-cancelled: employee deactivated".
5. System cancels all future confirmed desk bookings.
6. System adds comment "Unattended: assignee deactivated" to all open tasks.
7. System deactivates the linked user account (IsActive=false).
8. Historical records (leave, time logs, payroll, documents) remain stored in the system.
9. Action is recorded in the audit log.

US_10: Transfer employee to new department

1. HR Manager opens the employee profile and clicks Edit.
2. User selects a new department and optionally a new team.
3. System validates department and team availability.
4. User saves the changes.
5. System updates the employee department and creates a DepartmentTransferHistory record with old/new department, date, and reason.

6. If no explicit manager is set, system auto-assigns the new department head as manager.

US_11: Create department

1. HR Manager navigates to People > Departments tab.
2. User clicks New Department.
3. User enters department name, code (unique), and description.
4. User optionally selects a parent department for hierarchy.
5. System validates code uniqueness and creates the department.
6. Department appears in the overview and tree view.

US_12: Manage department head

1. HR Manager navigates to the Department Detail page (/departments/{slug}).
2. User clicks the Department Head metric card.
3. System opens a dialog showing the current head and a dropdown of department employees.
4. User selects a new head or chooses "None (remove head)".
5. User clicks Update Head.
6. System updates the department head and auto-assigns the head as manager for all department employees.

US_13: Create team

1. HR Manager navigates to a Department Detail page.
2. User clicks Add Team.
3. User enters team name, code (unique), optional description, and optionally selects a team lead from department employees.
4. System validates code uniqueness and creates the team.
5. If team lead is set, system auto-adds the lead as a team member.

US_14: Submit leave request

1. Employee navigates to Attendance > Leave.
2. Employee clicks Request Leave.
3. Employee selects leave type from configured types.
4. Employee chooses start date and end date.
5. Employee enters reason for leave.
6. If the leave type requires an attachment, employee uploads a supporting document.
7. System calculates business days (excluding weekends).
8. For paid leave types, system checks leave balance availability.
9. If valid, employee submits the request.
10. System creates the leave request with Pending status.
11. System sends notification to the employee manager and all HR Managers.

US_15: Recommend leave request

1. Team Lead or Department Manager navigates to the Team Requests tab under Leave.
2. System displays pending leave requests scoped to role: Team Lead sees direct reports only, Department Manager sees department employees.
3. Reviewer selects a request to review.
4. Reviewer checks employee name, leave type, dates, and reason.
5. Reviewer chooses Recommend or Reject and may add comments.
6. System updates the request status to Recommended or Rejected.
7. If recommended, the request is visible to HR for final approval.
8. Employee receives a notification with the recommender name: "[Name] has recommended your leave request."

US_16: Approve leave request

1. HR Manager navigates to the leave approval section.
2. System displays all Pending and Recommended leave requests across the organization.
3. HR selects a request to review.
4. HR checks request details, dates, and manager recommendation/comments.
5. HR chooses Approve or Reject and may add comments.
6. If approved, system deducts the approved days from the employee leave balance.
7. For unpaid leave, system automatically creates a salary deduction record.
8. Employee receives a notification with the final decision.

US_17: Cancel leave request

1. Employee navigates to Attendance > Leave and views their requests.
2. Employee selects a Pending, Recommended, or Approved request.
3. Employee clicks Cancel.
4. System updates the request status to Cancelled.
5. If the request was already Approved, system refunds the deducted leave balance.
6. If unpaid leave deduction was created, it is removed.
7. Response returns NoContent (204).

US_18: View leave balance

1. Employee navigates to Attendance > Leave or calls GET /api/leave-balances/me.
2. System retrieves the employee leave balance for the current year.
3. System displays total allocated days, used days, carried-over days, and remaining days for each leave type.
4. Employee reviews available leave balances.

US_19: Configure leave types

1. HR Manager navigates to Administration > Leave Config tab.

2. User creates or edits a leave type with: name, default days per year, paid/unpaid, allow carryover, max carryover days, require attachment, active/inactive.
3. System validates name uniqueness.
4. Changes take effect for new requests; existing requests are not affected.

US_20: Clock in

1. Employee navigates to Attendance > Time.
2. Employee clicks the Clock In button.
3. System checks for any open sessions from previous days.
4. If a stale session exists, system auto-closes it at StartTime + standard work hours with note "Auto-closed: employee did not clock out".
5. System records the current time as the start of a new session.
6. System updates employee status to Clocked In with a live timer.

US_21: Clock out

1. Employee navigates to Attendance > Time.
2. Employee clicks the Clock Out button.
3. Employee optionally adds notes.
4. System records the current time as the end of the session and calculates duration.
5. If the session exceeds standard daily hours, system auto-detects overtime and creates an OvertimeRecord.
6. System updates the attendance log.

US_22: View time logs

1. Employee navigates to Attendance > Time or calls GET /api/time-tracking/me.
2. System retrieves the employee daily and weekly time logs (default last 30 days).
3. System displays worked hours, sessions, and overtime indicators.
4. Employee can navigate between weeks to review previous records.
5. GET /api/time-tracking/me/summary returns the weekly summary.

US_23: Request time correction

1. Employee opens the time logs page.
2. Employee selects the incorrect time log entry.
3. Employee clicks Request Modification.
4. Employee enters corrected start and/or end time.
5. Employee provides a reason for the correction.
6. Employee submits the request.
7. System stores the request with Pending Approval status.
8. Manager receives a notification.

US_24: Approve time correction

1. Manager navigates to the Approvals tab under Time & Overtime.
2. System displays pending time correction requests scoped by role.
3. Manager selects a request to review.
4. Manager checks the original and corrected times and the reason.
5. Manager chooses Approve or Reject.
6. If approved, system updates the time log entry and checks for overlaps.
7. Employee receives a notification of the result.

US_25: Submit overtime request

1. Employee navigates to the overtime section.
2. Employee selects the date (must be today or past, not future).
3. Employee enters overtime minutes and reason.
4. Employee submits the overtime request.
5. System creates the request as Pending with type ManualRequest.
6. System sends notification to the employee manager and all HR Managers.

US_26: Approve overtime request

1. Reviewer navigates to the overtime approval section.
2. System displays pending overtime requests scoped by role.
3. Reviewer selects a request.
4. Reviewer checks date, hours, and justification.
5. Reviewer chooses Approve or Reject with optional comments.
6. System updates the overtime request status.
7. Employee is notified of the final decision.

US_27: Set employee salary

1. HR Manager navigates to Administration > Salary or to an employee profile > Salary tab.
2. User enters or updates base annual salary, currency, and effective date.
3. User can add recurring or one-time allowances (Transport, Housing, Meal, Custom).
4. User can add recurring or one-time bonuses with reason.
5. User can add deductions.
6. System saves the salary record.
7. System sends a notification to the employee: "Your salary has been updated. New rate effective from [date]."

US_28: View salary and payslip history

1. Employee navigates to Profile > My Salary.
2. System retrieves current salary details and projection (monthly base, daily rate, hourly rate).

3. System displays salary history with all past salary records and effective dates.
4. System displays payslip history (finalized payslips only).
5. Employee can click Download PDF on any finalized payslip.

US_29: Create payroll run

1. HR Manager navigates to Administration > Payroll.
2. User clicks Run Payroll.
3. User selects payroll month and year.
4. System validates no duplicate run exists for that month/year.
5. System checks all active employees have a salary record. If any are missing, system blocks the run and lists the employee names.
6. System logs a warning if any employee monthly salary exceeds $100,000.
7. System generates draft payslips for all active employees with calculated: base salary (annual/12), allowances, bonuses, deductions, gross pay, net pay, leave days, business days.
8. Each payslip includes line items tracking every component with source ID.

US_30: Edit payslip

1. HR Manager opens a draft payroll run.
2. HR selects a specific employee payslip.
3. HR reviews salary components and deductions.
4. HR adds, removes, or modifies line items such as bonuses or deductions.
5. System recalculates payslip totals automatically.
6. System updates the parent payroll run totals (TotalGross, TotalNet).
7. HR saves the changes.

US_31: Finalize payroll

1. HR Manager reviews all draft payslips in the run.
2. HR clicks Finalize Run.
3. System asks for confirmation because finalization is irreversible.
4. HR confirms the action.
5. System locks all payslips from further editing.
6. System changes payroll status to Finalized and records finalizer identity and timestamp.
7. Finalized payslips become visible to employees in their salary section.

US_32: Download payslip PDF

1. Employee navigates to Profile > My Salary.
2. System displays finalized payslip history.
3. Employee selects a payslip and clicks Download PDF.

4. System generates the PDF using QuestPDF with employee info, earnings table, deductions table, and net pay.
5. PDF is downloaded to the employee device.

US_33: Create task

1. User navigates to the Tasks module.
2. User clicks Create Task.
3. User enters task title and description.
4. User selects an employee to assign from a name dropdown (must be a manageable employee).
5. User selects priority (Low, Medium, High, Urgent) and due date.
6. User optionally sets a category.
7. User clicks Create.
8. System saves the task with status ToDo.
9. Assigned employee receives a notification.

US_34: Update task status

1. Employee opens the Tasks module.
2. Employee selects an assigned task.
3. Employee changes the task status (ToDo, InProgress, Done) via dropdown.
4. If status changes to Done, system sets CompletedAt timestamp.
5. If status changes away from Done, system clears CompletedAt.
6. System saves the updated status automatically.

US_35: Comment on task

1. User opens a task detail dialog.
2. User scrolls to the Comments section.
3. User writes a comment and clicks Send.
4. System stores the comment with timestamp and author identity.
5. Comment appears in the thread for all participants.

US_36: Upload document

1. HR Manager navigates to an employee profile > Documents tab or Administration > Documents.
2. User clicks Upload Document.
3. User selects a file (max 10MB; allowed: .pdf, .doc, .docx, .png, .jpg, .jpeg).
4. User selects a document category and optionally sets an expiry date and notes.
5. User confirms the upload.
6. System stores the file with a GUID-based filename in /uploads/documents/{employeeId}/.
7. Document is available to the employee and HR.

US_37: Download document

1. User navigates to the document list.
2. System displays available documents with name, category, upload date, and expiry.
3. User selects a document and clicks Download.
4. System retrieves the file and downloads it with the original filename.

US_38: Book a desk

1. Employee navigates to the Desk Booking module.
2. Employee selects an office from the dropdown.
3. Employee chooses a booking date (must be today or future, max 30 days ahead).
4. System displays available desks grouped by floor with label and amenities.
5. Employee selects a desk.
6. Employee optionally adds notes.
7. Employee clicks Confirm Booking.
8. System validates: date not in past, within 30-day horizon, desk is active, desk not already booked, employee has no other booking that day.
9. System creates the booking. It appears in My Bookings.

US_39: Cancel desk booking

1. Employee navigates to Desk Booking > My Bookings.
2. Employee selects a confirmed future booking.
3. Employee clicks Cancel.
4. System sets booking status to Cancelled.
5. The desk becomes available for others.

US_40: Manage offices, floors, and desks

1. HR Manager navigates to Desk Booking.
2. User creates offices with name and address.
3. User adds floors to an office with name and sort order.
4. User adds desks to a floor with label, amenities, and active status.
5. User can update or delete offices, floors, and desks.
6. Deleting an office cascades to soft-delete all child floors and desks.

US_41: Create announcement

1. HR Manager navigates to Announcements > Create tab.
2. User enters announcement title and content body.
3. User selects priority (Normal, Important, Urgent).
4. User optionally scopes to a specific department or leaves as company-wide.
5. User optionally toggles Pin.
6. User clicks Publish.

7. System creates the announcement and sends notifications to all relevant employees (excluding the author).

US_42: View and read announcements

1. User navigates to the Announcements page.
2. System displays announcements sorted by pinned first, then by publish date descending.
3. Unread announcements show a blue dot indicator.
4. User clicks an announcement to expand and read the full body.
5. System marks the announcement as read via the read receipt system.
6. Unread count in the page header updates.

US_43: Create onboarding template

1. HR Manager navigates to Administration > Onboarding.
2. User clicks Create Template.
3. User enters template name, description, and optionally targets a specific employment type.
4. User adds checklist items with description, responsible role (Employee, HR, Manager, IT), and default due days.
5. System saves the template for reuse across employees.

US_44: Assign onboarding checklist

1. HR Manager navigates to onboarding and selects an employee.
2. User selects a template to assign.
3. System creates an OnboardingChecklist with items and calculates due dates as assignment date + default due days.
4. Employee receives a notification about their onboarding checklist.

US_45: Complete onboarding item

1. Employee navigates to their onboarding checklist (via notification or profile).
2. Employee views all checklist items with descriptions, responsible roles, due dates, and status.
3. Employee marks an item as complete.
4. System sets CompletedAt timestamp and updates item status.
5. When all items are complete, checklist status changes to Completed.

US_46: Create and start performance review cycle

1. HR Manager navigates to People > Performance tab.
2. User clicks Create Cycle with name, start date, end date, and target scope (All or specific Department).
3. Cycle is created in Draft status.

4. User clicks Start Cycle.
5. System auto-creates a PerformanceReview for each in-scope employee with their current manager.
6. Employees without managers are skipped (logged).
7. Reviews are created with status PendingSelfAssessment.

US_47: Submit self-assessment

1. Employee is notified that a review cycle has started.
2. Employee navigates to Performance and opens their review.
3. Employee writes self-assessment text and provides a self-rating.
4. Employee submits the assessment.
5. Review status changes to PendingManagerReview.

US_48: Submit manager review

1. Manager views reviews for their direct reports or department.
2. Manager selects an employee review in PendingManagerReview status.
3. Manager writes review text, provides a rating, and sets an overall rating.
4. Manager submits the review.
5. Review status changes to PendingHRReview.

US_49: Set performance goals

1. Manager opens an employee performance review.
2. Manager adds goals with description, target date, and status (NotStarted, InProgress, Achieved, Missed).
3. Goals are linked to the review and visible to the employee.

US_50: Generate self-service PDF document

1. Employee navigates to My Documents page.
2. System displays three document cards: Employment Verification Letter, Salary Certificate, Leave Balance Statement.
3. Employee clicks Download PDF on the desired document.
4. System generates the PDF on-demand using QuestPDF with live data from the employee, salary, and leave balance records.
5. PDF is downloaded to the employee device.

US_51: Manage holidays

1. HR Manager navigates to Administration > Holidays.
2. User creates a holiday with name, date, description, and recurring flag.
3. System validates no duplicate holiday exists with same name and date.
4. User can edit or delete existing holidays.
5. Recurring holidays apply every year; one-time holidays apply to a specific date only.

US_52: View and manage notifications

1. User clicks the notification bell icon in the header (shows unread count badge).
2. System retrieves the user notifications with pagination.
3. System displays unread and read notifications with type icon, title, message, and relative timestamp.
4. User can filter between All and Unread only.
5. User clicks a notification to mark it as read.
6. User can click Mark All as Read to clear all unread notifications.

US_53: View calendar

1. User navigates to the Calendar page.
2. System displays a FullCalendar view with month/week/day toggles.
3. Events are color-coded: Tasks (indigo), Leave (green), Holidays (rose), Reviews (purple).
4. Tabs scope the view: My Calendar (all roles), Team (Team Lead+), Department (Department Manager+).
5. Events auto-load based on the visible date range.

US_54: View audit logs

1. HR Manager navigates to Administration > Audit Log tab or GET /api/audit-logs.
2. System displays a paginated list of audit entries sorted by timestamp descending.
3. Each entry shows: timestamp, user, action, entity type, entity ID, module, and details.
4. User can filter by module, action, user, entity type, and date range.
5. Audit logs are immutable and cannot be modified or deleted.

US_55: Manage user accounts

1. Super Admin navigates to Administration > Users tab.
2. System displays all user accounts with email, role, status, last login, and creation date.
3. Admin can create new user accounts with email, password, and role.
4. Admin can change a user role (Employee, Team Lead, Department Manager, HR Manager, Super Admin).
5. Admin can activate or deactivate user accounts. Deactivated accounts are blocked by ActiveUserMiddleware.
6. All account changes are recorded in the audit log.

US_56: Configure settings

1. User navigates to the Settings page.
2. User can switch between Light, Dark, and System themes via toggle buttons.
3. User can select a preferred language (English, Albanian, German) from the dropdown.

4. Language preference is saved to localStorage with confirmation toast: "Language preference saved. Translation support coming soon."

4.1.3 User Cases

UC-01: Login

UC Name:	Login
Dependency	US_01: User logs in
Actors	All Users
Preconditions	User has a valid account
Description of Main Sequence	1. User opens login page 2. User enters email and password 3. User clicks Sign In 4. System validates credentials 5. System generates JWT access and refresh tokens 6. System redirects to role-specific dashboard
Description of Alternative Sequence	a. Invalid credentials → error message "Invalid email or password" b. Account deactivated → access denied c. Account locked (5 failed attempts) → locked for 15 minutes d. Rate limit exceeded → HTTP 429 returned
Non-functional requirements	Security, Performance, Availability
Postconditions	User is logged in with valid session

UC-02: Logout

UC Name:	Logout
Dependency	US_02: User logs out
Actors	All Users
Preconditions	User is logged in
Description of Main Sequence	1. User opens profile menu 2. User clicks Logout 3. System revokes refresh token 4. System clears browser tokens 5. Redirects to login page
Description of Alternative Sequence	None
Non-functional requirements	Security, Reliability
Postconditions	User is logged out, tokens revoked

UC-03: Reset Password

UC Name:	Reset Password
Dependency	US_03: Reset password
Actors	All Users
Preconditions	User has a registered email
Description of Main Sequence	1. User clicks Forgot Password on login page 2. User enters email 3. System generates hashed reset token with 1-hour expiry 4. System returns generic success message (prevents email enumeration) 5. User receives reset link and sets new password 6. System updates password and revokes all refresh tokens
Description of Alternative Sequence	a. Invalid/expired token → error message b. Password too short → validation error
Non-functional requirements	Security, Usability
Postconditions	Password updated, all sessions invalidated

UC-04: Change Password

UC Name:	Change Password
Dependency	US_04: Change password
Actors	All Users
Preconditions	User is logged in
Description of Main Sequence	1. User navigates to Profile > Security 2. User enters current password and new password 3. System verifies current password 4. System updates password hash 5. System revokes all refresh tokens
Description of Alternative Sequence	a. Incorrect current password → error b. Weak new password → validation error
Non-functional requirements	Security, Encryption
Postconditions	Password updated successfully

UC-05: Create Employee

UC Name:	Create Employee
Dependency	US_05: Create employee
Actors	HR Manager, Super Admin
Preconditions	User is authenticated with HR/Admin role
Description of Main Sequence	1. User opens People section 2. Clicks Add Employee 3. Enters personal details (name, email, phone, DOB, address) 4. Enters employment details (department, team, job title, hire date, password) 5. Clicks Create

	6. System validates fields and email uniqueness 7. System creates employee record, user account, and leave balances
Description of Alternative Sequence	a. Email already exists → error b. Required fields missing → validation error
Non-functional requirements	Data Integrity, Security
Postconditions	Employee and user account created

UC-06: View Employee Profile

UC Name:	View Employee Profile
Dependency	US_06: View employee profile
Actors	All Roles (scoped)
Preconditions	User is logged in
Description of Main Sequence	1. User selects employee from directory 2. System retrieves employee details 3. System displays profile with personal info, employment, department, team, manager 4. HR/Admin sees additional Salary and Documents tabs
Description of Alternative Sequence	a. Employee not found → error b. Insufficient permissions → access denied
Non-functional requirements	Performance, Usability
Postconditions	Employee data displayed

UC-07: Update Employee

UC Name:	Update Employee
Dependency	US_08: Update employee profile
Actors	HR Manager, Super Admin
Preconditions	Employee exists
Description of Main Sequence	1. User opens employee profile and clicks Edit 2. User modifies details (department, job title, team, manager, status) 3. System validates updated information 4. If department changed, system creates transfer history record 5. If email changed, system syncs linked user account 6. User clicks Save
Description of Alternative Sequence	a. Invalid data → error b. Duplicate email → error
Non-functional requirements	Data Integrity, Security
Postconditions	Employee record updated

UC-08: Deactivate Employee

UC Name:	Deactivate Employee
Dependency	US_09: Deactivate employee
Actors	HR Manager, Super Admin
Preconditions	Employee exists and is active
Description of Main Sequence	1. User opens employee profile 2. User changes status to Terminated or Suspended 3. System confirms the action 4. System auto-cancels all pending/recommended leave requests 5. System cancels all future desk bookings 6. System adds "Unattended: assignee deactivated" comment to open tasks 7. System deactivates linked user account
Description of Alternative Sequence	a. Employee already inactive → no action
Non-functional requirements	Security, Reliability
Postconditions	Employee deactivated, cleanup completed

UC-09: Transfer Employee

UC Name:	Transfer Employee
Dependency	US_10: Transfer employee
Actors	HR Manager, Super Admin
Preconditions	Employee exists
Description of Main Sequence	1. User opens employee profile and clicks Edit 2. User selects new department and optionally new team 3. System validates department/team availability 4. User saves changes 5. System updates department and creates Department Transfer History record 6. System auto-assigns new department head as manager if no explicit manager set
Description of Alternative Sequence	a. Invalid department/team → error
Non-functional requirements	Data Integrity, Security
Postconditions	Employee transferred, history recorded

UC-10: View Employee Directory

UC Name:	View Employee Directory
Dependency	US_07: View employee directory
Actors	All Roles (scoped)
Preconditions	User is logged in

Description of Main Sequence	1. User navigates to People section 2. System retrieves employees scoped by role (own, team, department, or all) 3. System displays searchable table with Name, Department, Job Title, Status 4. User can search by name/email and filter by department, status, type
Description of Alternative Sequence	a. No employees found → empty state displayed
Non-functional requirements	Performance, Usability
Postconditions	Employee directory displayed

UC-11: Submit Leave Request

UC Name:	Submit Leave Request
Dependency	US_14: Submit leave request
Actors	Employee, Team Lead, Department Manager
Preconditions	User is logged in, leave balances initialized
Description of Main Sequence	1. User navigates to Attendance > Leave 2. Clicks Request Leave 3. Selects leave type, start date, end date 4. Enters reason 5. If required, uploads supporting document 6. System calculates business days and checks balance 7. System creates request with Pending status 8. System notifies employee manager and all HR Managers
Description of Alternative Sequence	a. Insufficient balance → error with available/requested days b. No manager and no department assigned → request rejected c. End date before start date → validation error
Non-functional requirements	Reliability, Accuracy
Postconditions	Leave request submitted with Pending status

UC-12: Recommend Leave Request

UC Name:	Recommend Leave Request
Dependency	US_15: Recommend leave request
Actors	Team Lead, Department Manager
Preconditions	Pending leave request exists from a managed employee
Description of Main Sequence	1. Reviewer navigates to Team Requests tab 2. System displays pending requests scoped by role 3. Reviewer selects a request and reviews details 4. Reviewer chooses Recommend or Reject with optional comments

	5. System updates status to Recommended or Rejected 6. Employee receives notification with recommender name
Description of Alternative Sequence	a. Target employee is HR staff → only SuperAdmin can process b. Request already processed → no action
Non-functional requirements	Security, Consistency
Postconditions	Request recommended and routed to HR for final approval

UC-13: Approve Leave Request (Tier 2)

UC Name:	Approve Leave Request
Dependency	US_16: Approve leave request
Actors	HR Manager, Super Admin
Preconditions	Leave request exists in Pending or Recommended status
Description of Main Sequence	1. HR navigates to leave approval section 2. System displays all Pending and Recommended requests 3. HR selects a request and reviews details and recommendation 4. HR chooses Approve or Reject with optional comments 5. If approved, system deducts leave balance 6. For unpaid leave, system creates salary deduction record 7. Employee receives notification of final decision
Description of Alternative Sequence	a. Already processed → no action b. Cross-year leave → days split by year and deducted independently
Non-functional requirements	Consistency, Security
Postconditions	Request finalized, balance updated if approved

UC-14: Cancel Leave Request

UC Name:	Cancel Leave Request
Dependency	US_17: Cancel leave request
Actors	Employee (own request)
Preconditions	Leave request exists in Pending, Recommended, or Approved status
Description of Main Sequence	1. User navigates to Leave page 2. Selects the request 3. Clicks Cancel 4. System updates status to Cancelled 5. If request was Approved, system refunds deducted balance 6. If unpaid leave deduction was created, it is removed
Description of Alternative Sequence	a. Request already processed/cancelled → cancellation denied
Non-functional requirements	Consistency, Reliability
Postconditions	Leave request cancelled, balance restored if applicable

UC-15: View Leave Balance

UC Name:	View Leave Balance
Dependency	US_18: View leave balance
Actors	All Roles (own balance)
Preconditions	User is logged in
Description of Main Sequence	1. User navigates to Attendance > Leave or calls /api/leave-balances/me 2. System retrieves balances for current year 3. System displays total, used, carried-over, and remaining days per leave type
Description of Alternative Sequence	a. No balances initialized → empty list returned
Non-functional requirements	Performance, Accuracy
Postconditions	Leave balances displayed

UC-16: Configure Leave Types

UC Name:	Configure Leave Types
Dependency	US_19: Configure leave types
Actors	HR Manager, Super Admin
Preconditions	User is authenticated with HR/Admin role
Description of Main Sequence	1. HR navigates to Administration > Leave Config 2. Creates or edits leave type with name, days/year, paid/unpaid, carryover rules, attachment required 3. System validates name uniqueness 4. Changes take effect for new requests
Description of Alternative Sequence	a. Duplicate name → error
Non-functional requirements	Data Integrity, Security
Postconditions	Leave type configured

UC-17: Clock In

UC Name:	Clock In
Dependency	US_20: Clock in
Actors	Employee, Team Lead, Department Manager
Preconditions	User is logged in
Description of Main Sequence	1. User navigates to Attendance > Time 2. Clicks Clock In 3. System checks for stale sessions from previous days 4. If stale session found, auto-closes at standard hours with note "Auto-closed" 5. System records start time for new session 6. System displays live session timer

Description of Alternative Sequence	a. Active session already exists for today → error
Non-functional requirements	Accuracy, Performance
Postconditions	Session started

UC-18: Clock Out

UC Name:	Clock Out
Dependency	US_21: Clock out
Actors	Employee, Team Lead, Department Manager
Preconditions	Active session exists
Description of Main Sequence	1. User clicks Clock Out 2. Optionally adds notes 3. System records end time and calculates duration 4. If hours exceed standard, system auto-detects overtime 5. System updates attendance log
Description of Alternative Sequence	a. Session exceeds 16 hours → auto-capped at standard hours with adjustment note b. No active session → error
Non-functional requirements	Accuracy, Performance
Postconditions	Session ended, duration recorded

UC-19: Request Time Correction

UC Name:	Request Time Correction
Dependency	US_23: Request time correction
Actors	Employee, Team Lead, Department Manager
Preconditions	Time log entry exists
Description of Main Sequence	1. User selects incorrect time log 2. Clicks Request Modification 3. Enters corrected start/end time and reason 4. Submits request 5. System stores request as Pending 6. Manager receives notification
Description of Alternative Sequence	a. Same-day edits allowed directly without modification request
Non-functional requirements	Accuracy, Auditability
Postconditions	Modification request submitted

UC-20: Approve Time Correction

UC Name:	Approve Time Correction
Dependency	US_24: Approve time correction
Actors	Team Lead, Department Manager, HR Manager, Super Admin
Preconditions	Pending time correction request exists

Description of Main Sequence	1. Reviewer navigates to Approvals tab 2. Selects request and reviews original vs corrected times 3. Chooses Approve or Reject 4. If approved, system updates time log and checks for overlaps 5. Employee receives notification
Description of Alternative Sequence	a. Overlapping sessions after correction → rejection suggested
Non-functional requirements	Accuracy, Consistency
Postconditions	Time log corrected or request rejected

UC-21: Submit Overtime Request

UC Name:	Submit Overtime Request
Dependency	US_25: Submit overtime request
Actors	Employee, Team Lead, Department Manager
Preconditions	User is logged in, date is today or past
Description of Main Sequence	1. User navigates to overtime section 2. Selects date (today or past, not future) 3. Enters overtime minutes and reason 4. Submits request 5. System creates request as Pending with type ManualRequest 6. System notifies manager and all HR Managers
Description of Alternative Sequence	a. Future date → error b. Employee not Active/Probation → error
Non-functional requirements	Reliability, Accuracy
Postconditions	Overtime request submitted

UC-22: Approve Overtime Request

UC Name:	Approve Overtime Request
Dependency	US_26: Approve overtime request
Actors	Team Lead, Department Manager, HR Manager, Super Admin
Preconditions	Pending overtime request exists
Description of Main Sequence	1. Reviewer navigates to overtime approval section 2. Selects request and reviews date, hours, justification 3. Chooses Approve or Reject with optional comments 4. System updates status 5. Employee notified
Description of Alternative Sequence	a. HR staff overtime → only SuperAdmin can approve
Non-functional requirements	Security, Consistency
Postconditions	Overtime request processed

UC-23: Set Employee Salary

UC Name:	Set Employee Salary
Dependency	US_27: Set employee salary
Actors	HR Manager, Super Admin
Preconditions	Employee exists
Description of Main Sequence	1. HR navigates to employee Salary tab or Administration > Salary 2. Enters base annual salary, currency, effective date 3. Optionally adds allowances (Transport, Housing, Meal, Custom) 4. Optionally adds bonuses with reason 5. Optionally adds deductions 6. System saves and notifies employee: "Your salary has been updated"
Description of Alternative Sequence	a. Employee not found → error
Non-functional requirements	Data Integrity, Security
Postconditions	Salary record created, employee notified

UC-24: View Salary History

UC Name:	View Salary History
Dependency	US_28: View salary history
Actors	Employee (own)
Preconditions	User is logged in
Description of Main Sequence	1. Employee navigates to Profile > My Salary 2. System retrieves current salary, projection, and history 3. System displays base salary, allowances, bonuses, deductions 4. System shows payslip history (finalized only)
Description of Alternative Sequence	a. No salary record → empty state
Non-functional requirements	Performance, Accuracy
Postconditions	Salary details displayed

UC-25: Create Payroll Run

UC Name:	Create Payroll Run
Dependency	US_29: Create payroll run
Actors	HR Manager, Super Admin
Preconditions	Salary data exists for employees
Description of Main Sequence	1. HR navigates to Administration > Payroll 2. Clicks Run Payroll 3. Selects month and year

Description of Main Sequence	4. System validates no duplicate run exists 5. System checks all active employees have salary records (blocks with names if not) 6. System warns if any monthly salary exceeds $100,000 7. System generates draft payslips with calculated base, allowances, bonuses, deductions, net pay
Description of Alternative Sequence	a. Duplicate month/year → error b. Missing salary records → error listing employee names
Non-functional requirements	Performance, Accuracy
Postconditions	Draft payroll run created with payslips

UC-26: Edit Payslip

UC Name:	Edit Payslip
Dependency	US_30: Edit payslip
Actors	HR Manager, Super Admin
Preconditions	Draft payroll run exists
Description of Main Sequence	1. HR opens draft payroll run 2. Selects employee payslip 3. Reviews and modifies line items (bonuses, deductions, notes) 4. System recalculates totals automatically 5. System updates parent run totals 6. HR saves changes
Description of Alternative Sequence	a. Payroll already finalized → edit blocked
Non-functional requirements	Data Integrity, Accuracy
Postconditions	Payslip updated, run totals recalculated

UC-27: Finalize Payroll

UC Name:	Finalize Payroll
Dependency	US_31: Finalize payroll
Actors	HR Manager, Super Admin
Preconditions	Draft payroll run exists with reviewed payslips
Description of Main Sequence	1. HR reviews all payslips 2. Clicks Finalize Run 3. System asks for confirmation (irreversible) 4. HR confirms 5. System locks all payslips 6. Status changes to Finalized with timestamp and finalizer identity 7. Employees can now view their payslips
Description of Alternative Sequence	a. Missing data → system prevents finalization

Non-functional requirements	Security, Integrity, Performance
Postconditions	Payroll finalized and locked

UC-28: Download Payslip PDF

UC Name:	Download Payslip PDF
Dependency	US_32: Download payslip PDF
Actors	Employee (own, finalized); HR Manager, Super Admin (any)
Preconditions	Finalized payslip exists
Description of Main Sequence	1. User navigates to salary section 2. Selects a payslip 3. Clicks Download PDF 4. System generates PDF with QuestPDF (employee info, earnings, deductions, net pay) 5. PDF downloaded to device
Description of Alternative Sequence	a. Payslip not finalized → employee cannot access b. Employee tries to download another employee payslip → access denied
Non-functional requirements	Security, Performance
Postconditions	Payslip PDF downloaded

UC-29: Create Task

UC Name:	Create Task
Dependency	US_33: Create task
Actors	Team Lead, Department Manager, HR Manager, Super Admin (scoped)
Preconditions	User is logged in
Description of Main Sequence	1. User navigates to Tasks module 2. Clicks Create Task 3. Enters title, description, selects assignee from name dropdown 4. Selects priority (Low/Medium/High/Urgent) and due date 5. Clicks Create 6. System saves task with status ToDo 7. Assigned employee receives notification
Description of Alternative Sequence	a. Cannot manage assignee → 403 Forbidden b. Invalid input → validation error
Non-functional requirements	Usability, Performance
Postconditions	Task created and assigned

UC-30: Update Task Status

UC Name:	Update Task Status
Dependency	US_34: Update task status

Actors	Assigned employee, creator, managers
Preconditions	Task exists
Description of Main Sequence	1. User opens Tasks module 2. Selects task 3. Changes status via dropdown (ToDo → InProgress → Done) 4. System saves status automatically 5. If Done, system sets CompletedAt timestamp
Description of Alternative Sequence	a. Invalid status transition → error
Non-functional requirements	Usability
Postconditions	Task status updated

UC-31: Comment on Task

UC Name:	Comment on Task
Dependency	US_35: Comment on task
Actors	All users with task access
Preconditions	Task exists
Description of Main Sequence	1. User opens task detail 2. Scrolls to Comments section 3. Writes comment and clicks Send 4. System stores comment with timestamp and author 5. Comment visible to all task participants
Description of Alternative Sequence	a. Empty comment → validation error
Non-functional requirements	Usability
Postconditions	Comment added to task

UC-32: Upload Document

UC Name:	Upload Document
Dependency	US_36: Upload document
Actors	HR Manager, Super Admin
Preconditions	User is authenticated
Description of Main Sequence	1. User navigates to employee Documents tab 2. Clicks Upload 3. Selects file (max 10MB; .pdf, .doc, .docx, .png, .jpg, .jpeg) 4. Selects category and optionally sets expiry date and notes 5. Confirms upload 6. System stores file with GUID filename in /uploads/documents/{employeeId}/
Description of Alternative Sequence	a. Invalid file type or size → error
Non-functional requirements	Security, Storage Efficiency

Postconditions	Document saved and accessible

UC-33: Download Document

UC Name:	Download Document
Dependency	US_37: Download document
Actors	Employee (own); HR Manager, Super Admin (any)
Preconditions	Document exists
Description of Main Sequence	1. User navigates to document list 2. Selects document 3. Clicks Download 4. System retrieves and returns file
Description of Alternative Sequence	a. Not authorized → access denied
Non-functional requirements	Security, Performance
Postconditions	Document downloaded

UC-34: Generate Self-Service PDF

UC Name:	Generate Self-Service PDF
Dependency	US_50: Generate self-service PDF
Actors	All Roles
Preconditions	User is logged in with active employee profile
Description of Main Sequence	1. User navigates to My Documents page 2. Selects document type: Employment Verification, Salary Certificate, or Leave Balance 3. Clicks Download PDF 4. System generates PDF on demand from live data using QuestPDF 5. PDF downloaded to device
Description of Alternative Sequence	a. No salary record → Salary Certificate returns error b. No leave balances → Leave Balance returns error
Non-functional requirements	Performance, Accuracy
Postconditions	PDF document generated and downloaded

UC-35: Book a Desk

UC Name:	Book a Desk
Dependency	US_38: Book a desk
Actors	All Roles
Preconditions	User is logged in, offices/desks exist
Description of Main Sequence	1. User navigates to Desk Booking 2. Selects office and date 3. System displays available desks grouped by floor 4. User selects a desk and optionally adds notes 5. Clicks Confirm Booking

Description of Main Sequence	6. System validates: not past date, within 30 days, desk active, no double booking, one per employee per day 7. Booking created and appears in My Bookings
Description of Alternative Sequence	a. Past date → "Cannot book a desk for a past date" b. Beyond 30 days → "Cannot book more than 30 days in advance" c. Desk already booked → error d. Employee already has booking that day → error
Non-functional requirements	Reliability, Usability
Postconditions	Desk booked successfully

UC-36: Cancel Desk Booking

UC Name:	Cancel Desk Booking
Dependency	US_39: Cancel desk booking
Actors	Employee (own); HR Manager, Super Admin (any)
Preconditions	Confirmed booking exists
Description of Main Sequence	1. User navigates to My Bookings 2. Selects booking 3. Clicks Cancel 4. System sets status to Cancelled 5. Desk becomes available for others
Description of Alternative Sequence	a. Non-owner without HR role → access denied
Non-functional requirements	Reliability, Consistency
Postconditions	Booking cancelled, desk freed

UC-37: Manage Offices and Desks

UC Name:	Manage Offices and Desks
Dependency	US_40: Manage offices, floors, desks
Actors	HR Manager, Super Admin
Preconditions	User is authenticated with admin role
Description of Main Sequence	1. User creates offices with name and address 2. Adds floors with name and sort order 3. Adds desks with label, amenities, and active status 4. Can update or delete any entity 5. Deleting office cascades to soft-delete floors and desks
Description of Alternative Sequence	a. Duplicate data → validation error
Non-functional requirements	Security, Data Integrity
Postconditions	Office structure managed

UC-38: Create Announcement

UC Name:	Create Announcement
Dependency	US_41: Create announcement
Actors	HR Manager, Super Admin
Preconditions	User is logged in
Description of Main Sequence	1. User navigates to Announcements > Create tab 2. Enters title and content body 3. Selects priority (Normal/Important/Urgent) 4. Optionally scopes to specific department 5. Optionally toggles Pin 6. Clicks Publish 7. System creates announcement and notifies relevant employees (excluding author)
Description of Alternative Sequence	a. Missing content → validation error
Non-functional requirements	Availability, Performance
Postconditions	Announcement published

UC-39: View Notifications

UC Name:	View Notifications
Dependency	US_52: View and manage notifications
Actors	All Roles
Preconditions	User is logged in
Description of Main Sequence	1. User clicks notification bell or navigates to Notifications page 2. System retrieves notifications with pagination 3. Displays unread and read notifications with type icon, title, message, timestamp 4. User can filter All/Unread 5. User clicks notification to mark as read 6. User can click Mark All as Read
Description of Alternative Sequence	a. No notifications → empty state
Non-functional requirements	Performance, Usability
Postconditions	Notifications viewed and managed

UC-40: Create Onboarding Template

UC Name:	Create Onboarding Template
Dependency	US_43: Create onboarding template
Actors	HR Manager, Super Admin
Preconditions	User is authenticated
Description of Main Sequence	1. HR navigates to Administration > Onboarding 2. Clicks Create Template

Description of Main Sequence	3. Enters name, description, optional target employment type 4. Adds checklist items with description, responsible role (Employee/HR/Manager/IT), and default due days 5. Saves template for reuse
Description of Alternative Sequence	a. Missing required fields → validation error
Non-functional requirements	Usability, Data Integrity
Postconditions	Onboarding template created

UC-41: Assign Onboarding Checklist

UC Name:	Assign Onboarding Checklist
Dependency	US_44: Assign onboarding checklist
Actors	HR Manager, Super Admin
Preconditions	Template and employee exist
Description of Main Sequence	1. HR selects employee and template 2. System creates checklist with items 3. Due dates calculated as assignment date + default due days 4. Employee receives notification about their onboarding
Description of Alternative Sequence	a. Template not found → error
Non-functional requirements	Reliability
Postconditions	Onboarding checklist assigned to employee

UC-42: Complete Onboarding Item

UC Name:	Complete Onboarding Item
Dependency	US_45: Complete onboarding item
Actors	All Roles
Preconditions	Onboarding checklist assigned to employee
Description of Main Sequence	1. Employee views onboarding checklist 2. Selects an item and marks it as complete 3. System sets Completed At timestamp 4. When all items complete, checklist status changes to Completed
Description of Alternative Sequence	a. Item already completed → no action
Non-functional requirements	Usability
Postconditions	Onboarding item completed

UC-43: Create Performance Review Cycle

UC Name:	Create Performance Review Cycle
Dependency	US_46: Create review cycle
Actors	HR Manager, Super Admin

Preconditions	User is authenticated
Description of Main Sequence	1. HR navigates to People > Performance 2. Clicks Create Cycle with name, start/end dates, target scope (All or Department) 3. Cycle created in Draft status 4. HR clicks Start Cycle 5. System auto-creates reviews for in-scope employees with their current manager 6. Employees without managers are skipped
Description of Alternative Sequence	a. No employees in scope → empty cycle
Non-functional requirements	Data Integrity, Performance
Postconditions	Review cycle created and reviews assigned

UC-44: Submit Self-Assessment

UC Name:	Submit Self-Assessment
Dependency	US_47: Submit self-assessment
Actors	Employee
Preconditions	Active review cycle exists with a review for this employee
Description of Main Sequence	1. Employee navigates to Performance and opens their review 2. Writes self-assessment text and provides self-rating 3. Submits assessment 4. Review status changes to PendingManagerReview
Description of Alternative Sequence	a. Review not in PendingSelfAssessment status → blocked
Non-functional requirements	Usability
Postconditions	Self-assessment submitted

UC-45: Submit Manager Review

UC Name:	Submit Manager Review
Dependency	US_48: Submit manager review
Actors	Team Lead, Department Manager
Preconditions	Review in PendingManagerReview status
Description of Main Sequence	1. Manager opens employee review 2. Writes review text, provides rating and overall rating 3. Submits review 4. Review status changes to PendingHRReview
Description of Alternative Sequence	a. Not the assigned manager → access denied
Non-functional requirements	Security, Integrity
Postconditions	Manager review submitted

UC-46: Create Department

UC Name:	Create Department
Dependency	US_11: Create department
Actors	HR Manager, Super Admin
Preconditions	User is logged in with admin permissions
Description of Main Sequence	1. User navigates to Departments section 2. Clicks Create Department 3. Enters name, code (unique), and description 4. Optionally selects parent department 5. Clicks Save 6. System validates code uniqueness and creates department
Description of Alternative Sequence	a. Duplicate code → error b. Missing required fields → error
Non-functional requirements	Security, Data Validation, Performance
Postconditions	Department created

UC-47: Update Department / Manage Head

UC Name:	Update Department / Manage Head
Dependency	US_12: Manage department head
Actors	HR Manager, Super Admin
Preconditions	Department exists
Description of Main Sequence	1. User navigates to department detail page 2. Clicks Department Head metric card 3. Selects new head from department employee dropdown or removes head 4. System updates head and auto-assigns as manager for department employees
Description of Alternative Sequence	a. Invalid data → system rejects update
Non-functional requirements	Data Integrity, Security
Postconditions	Department updated, head assigned

UC-48: View Department Hierarchy

UC Name:	View Department Hierarchy
Dependency	US_11: Create department
Actors	HR Manager, Department Manager
Preconditions	Departments exist
Description of Main Sequence	1. User opens department module 2. System displays hierarchical tree structure 3. User clicks department to view detail page with teams and employees
Description of Alternative Sequence	a. No departments → empty state

Non-functional requirements	Usability, Performance
Postconditions	Organizational structure displayed

UC-49: Manage User Accounts

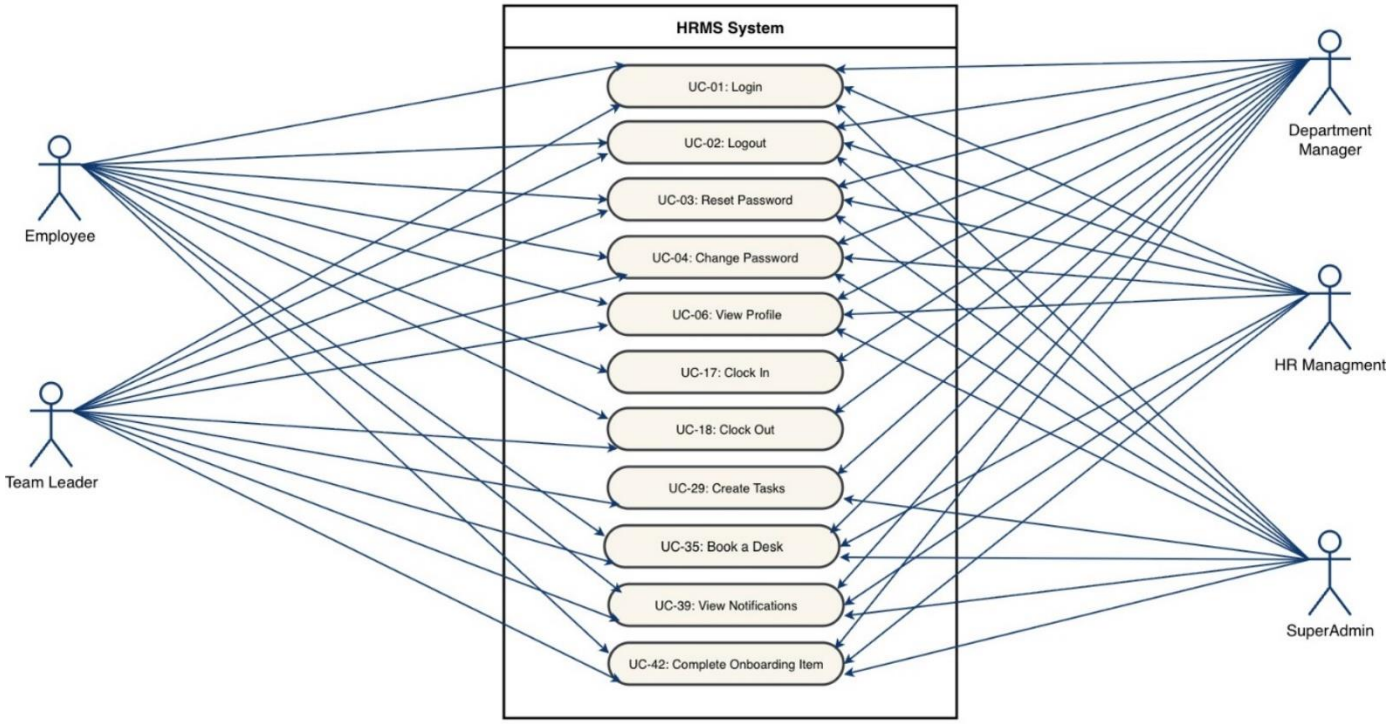
UC Name:	Manage User Accounts
Dependency	US_55: Manage user accounts
Actors	Super Admin
Preconditions	Admin is logged in
Description of Main Sequence	1. Admin navigates to Administration > Users 2. Views all accounts with email, role, status, last login 3. Can create new accounts, change roles, activate/deactivate 4. All changes recorded in audit log
Description of Alternative Sequence	a. Invalid data → error b. Deactivated accounts blocked by middleware
Non-functional requirements	Security, Integrity
Postconditions	User accounts managed

UC-50: Manage Holidays

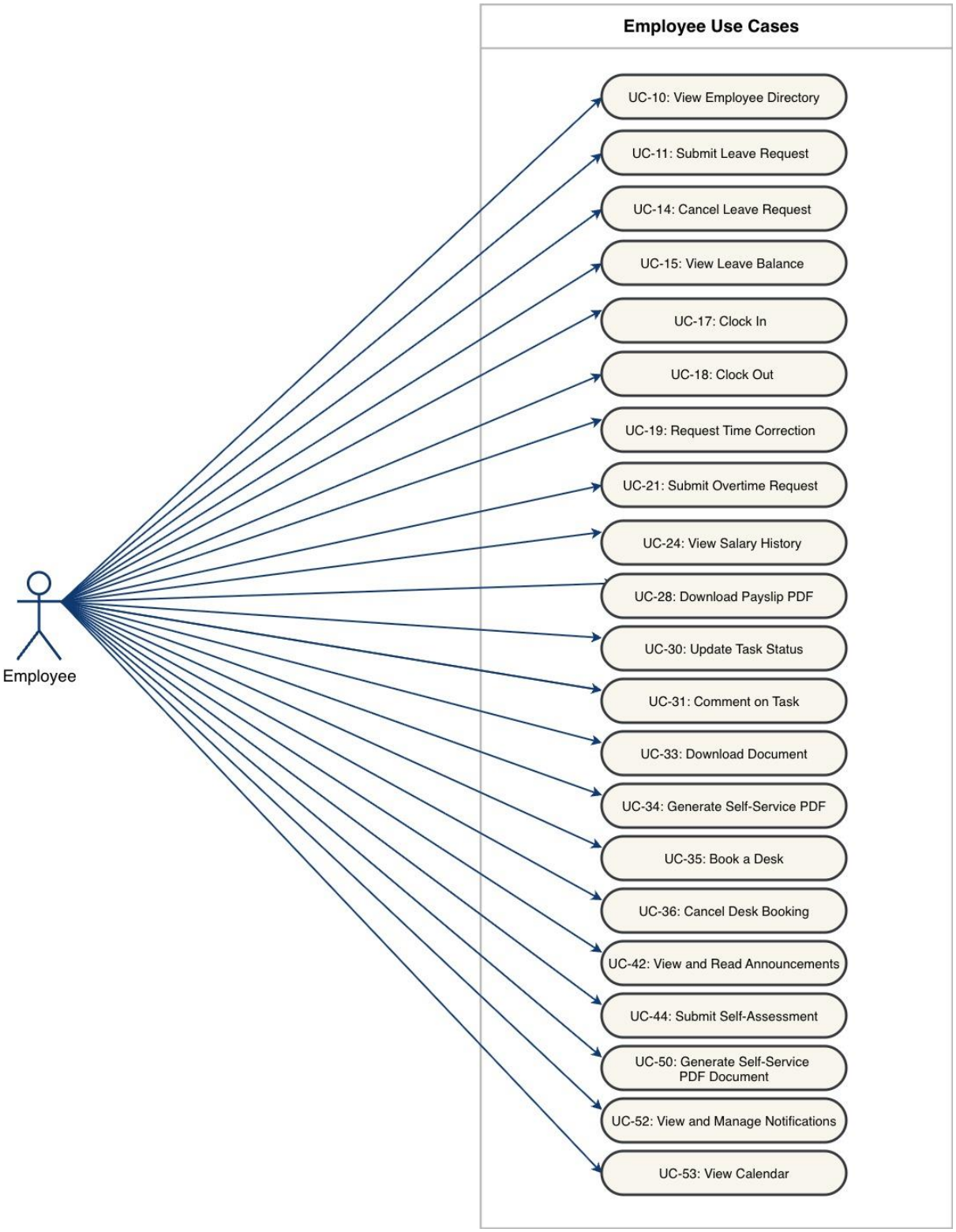
UC Name:	Manage Holidays
Dependency	US_51: Manage holidays
Actors	HR Manager, Super Admin
Preconditions	User is logged in
Description of Main Sequence	1. HR navigates to Administration > Holidays 2. Creates holiday with name, date, description, and recurring flag 3. System validates no duplicate (same name + date) 4. User can edit or delete holidays 5. Recurring holidays apply every year
Description of Alternative Sequence	a. Duplicate name + date → error
Non-functional requirements	Data Integrity
Postconditions	Holidays managed

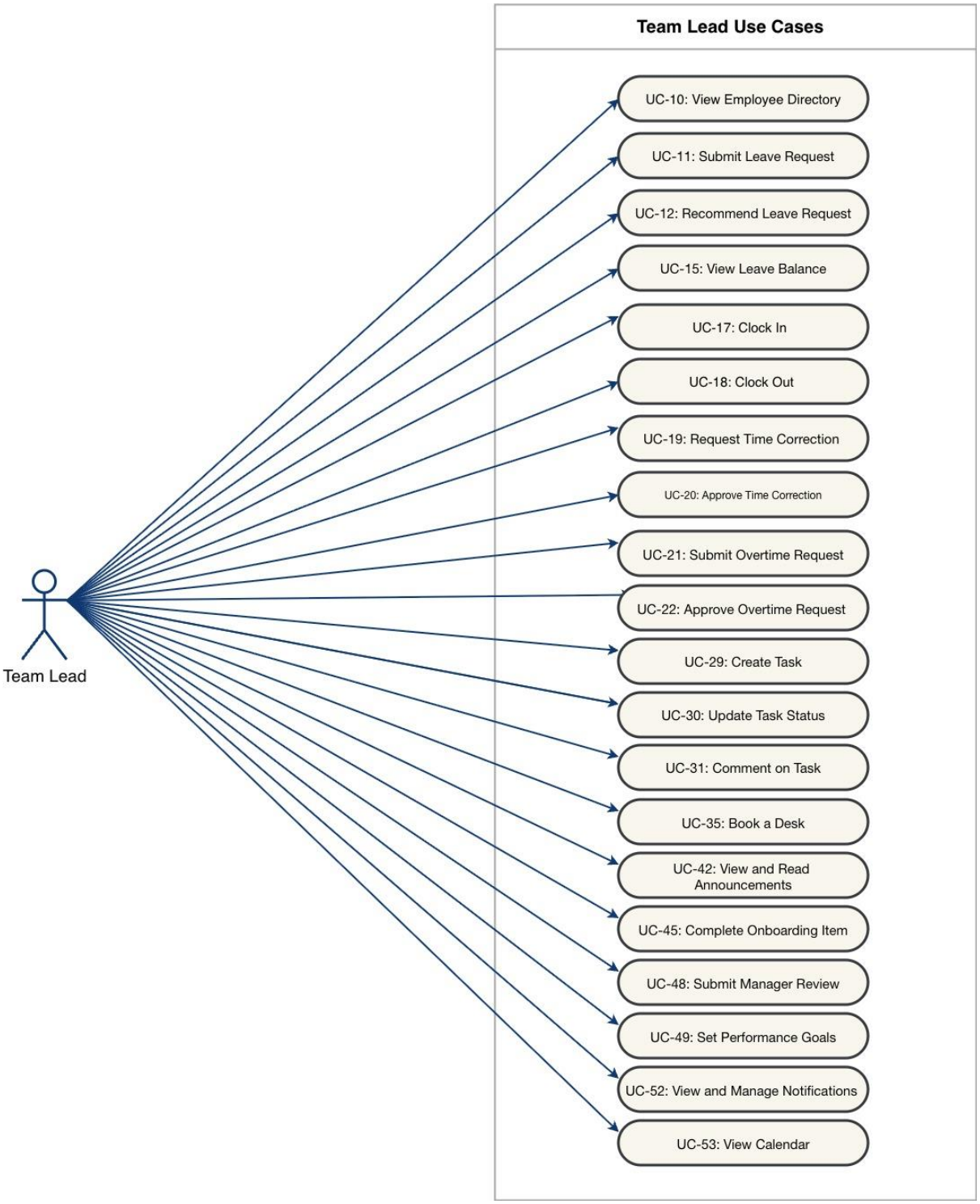
4.2 Behavioural Diagrams

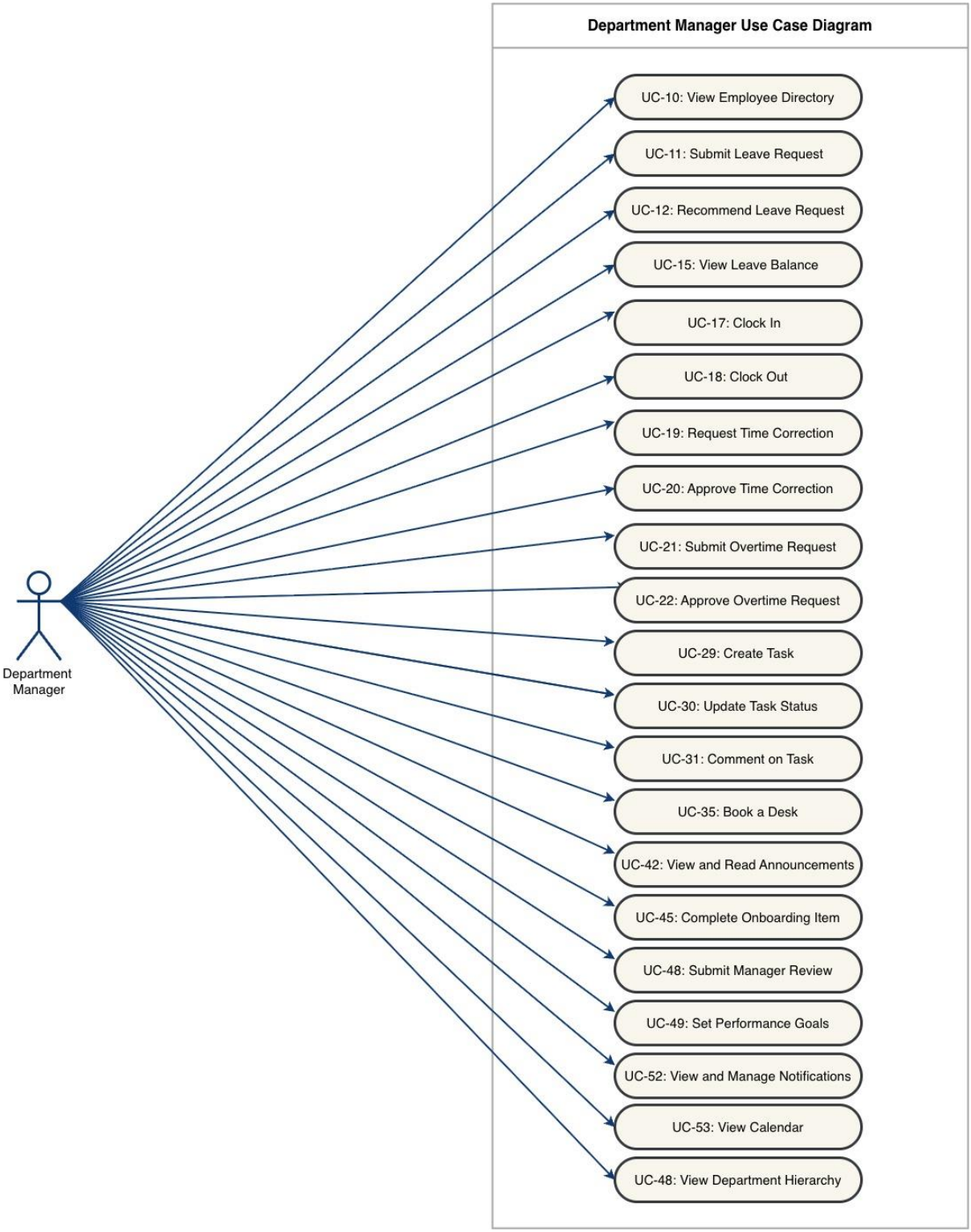
4.2.1 Use Case Diagrams

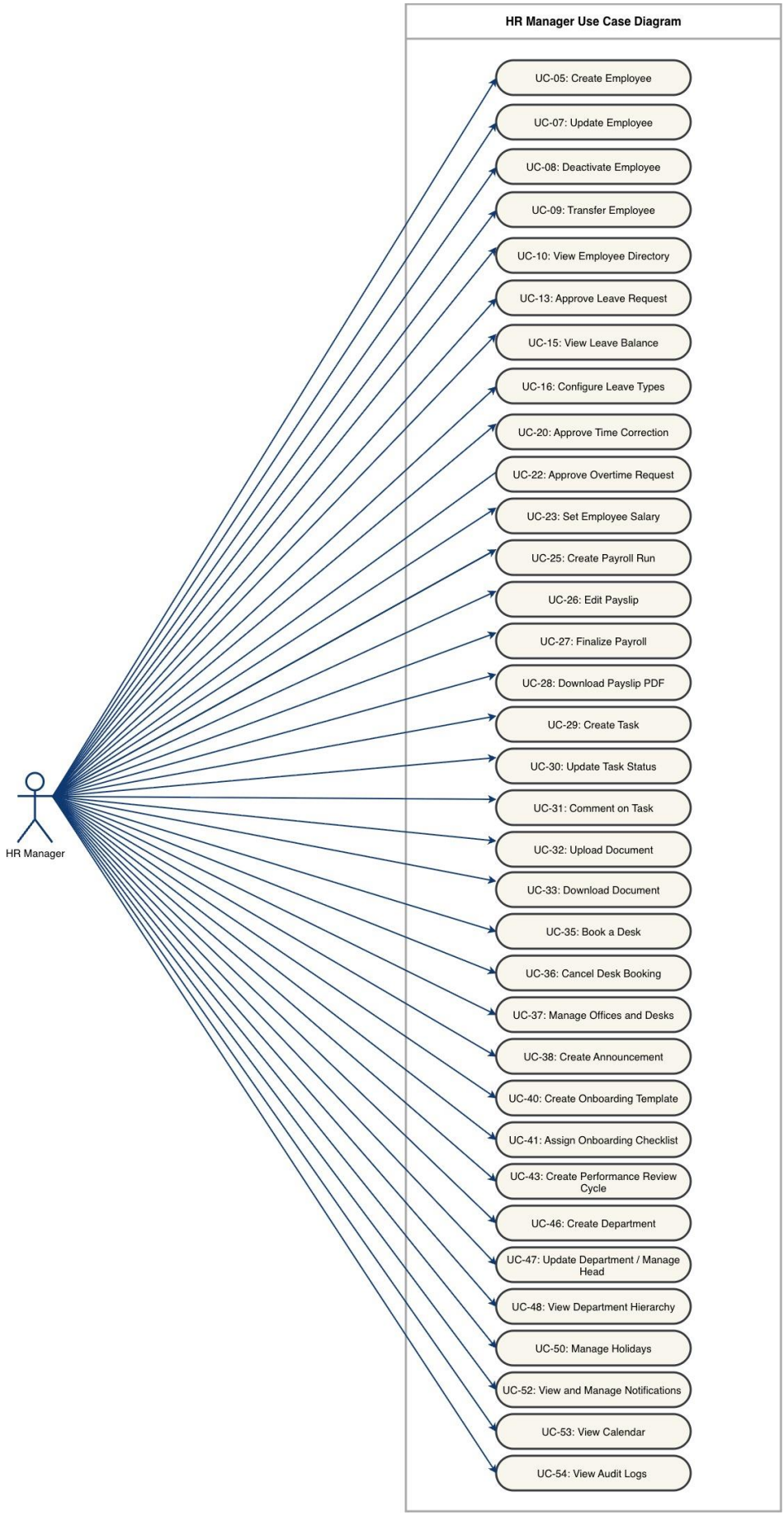


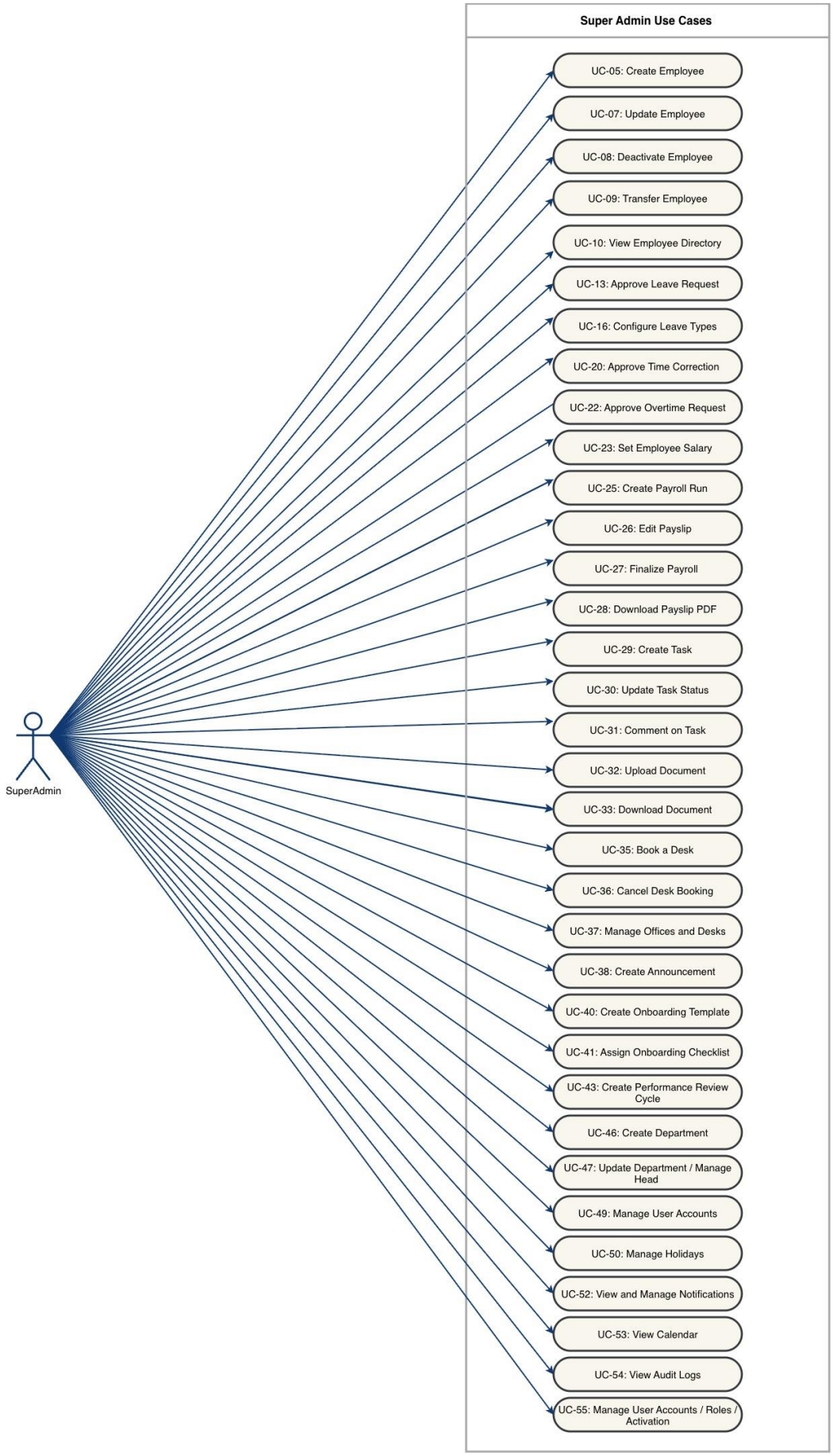
HRMS Overall Use Case Diagram

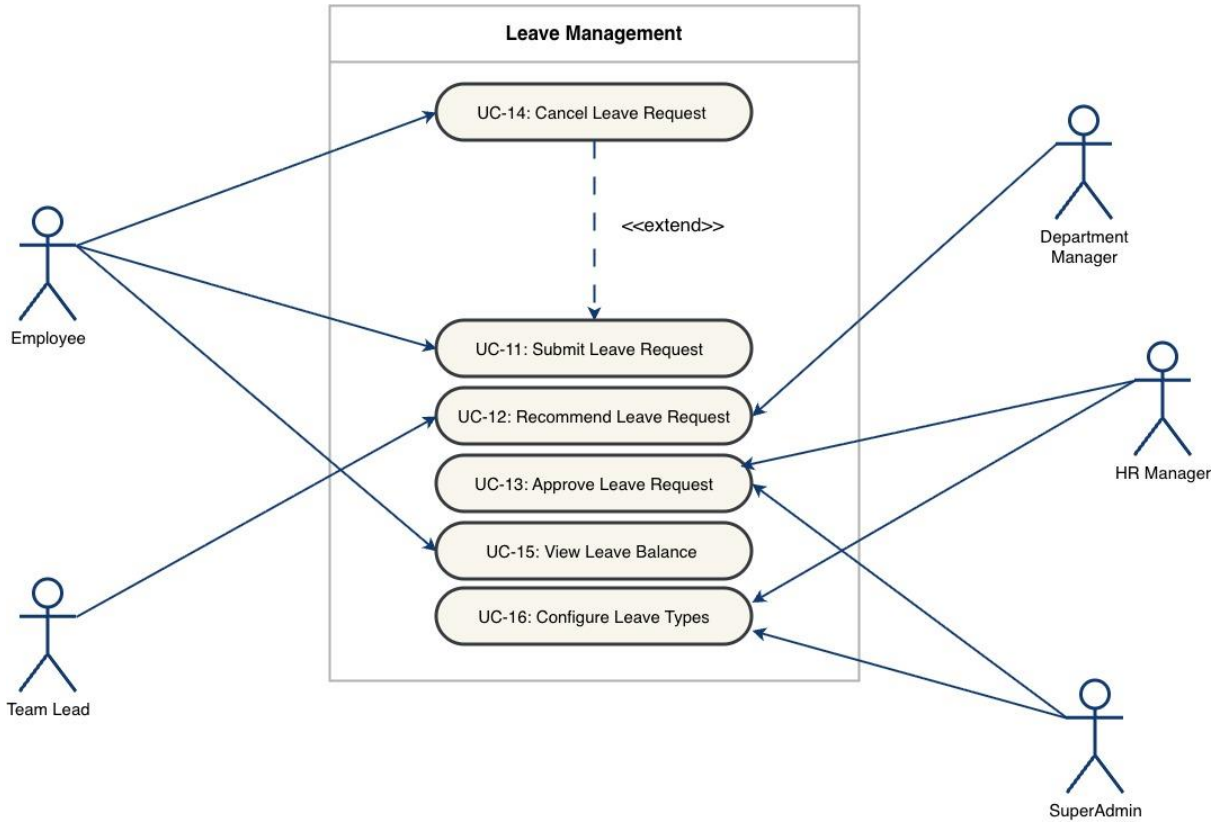

Employee Use Cases
Employee
UC-10: View Employee Directory
UC-11: Submit Leave Request
UC-14: Cancel Leave Request
UC-15: View Leave Balance
UC-17: Clock In
UC-18: Clock Out
UC-19: Request Time Correction
UC-21: Submit Overtime Request
UC-24: View Salary History
UC-28: Download Payslip PDF
UC-30: Update Task Status
UC-31: Comment on Task
UC-33: Download Document
UC-34: Generate Self-Service PDF
UC-35: Book a Desk
UC-36: Cancel Desk Booking
UC-42: View and Read Announcements
UC-44: Submit Self-Assessment
UC-50: Generate Self-Service PDF Document
UC-52: View and Manage Notifications
UC-53: View Calendar

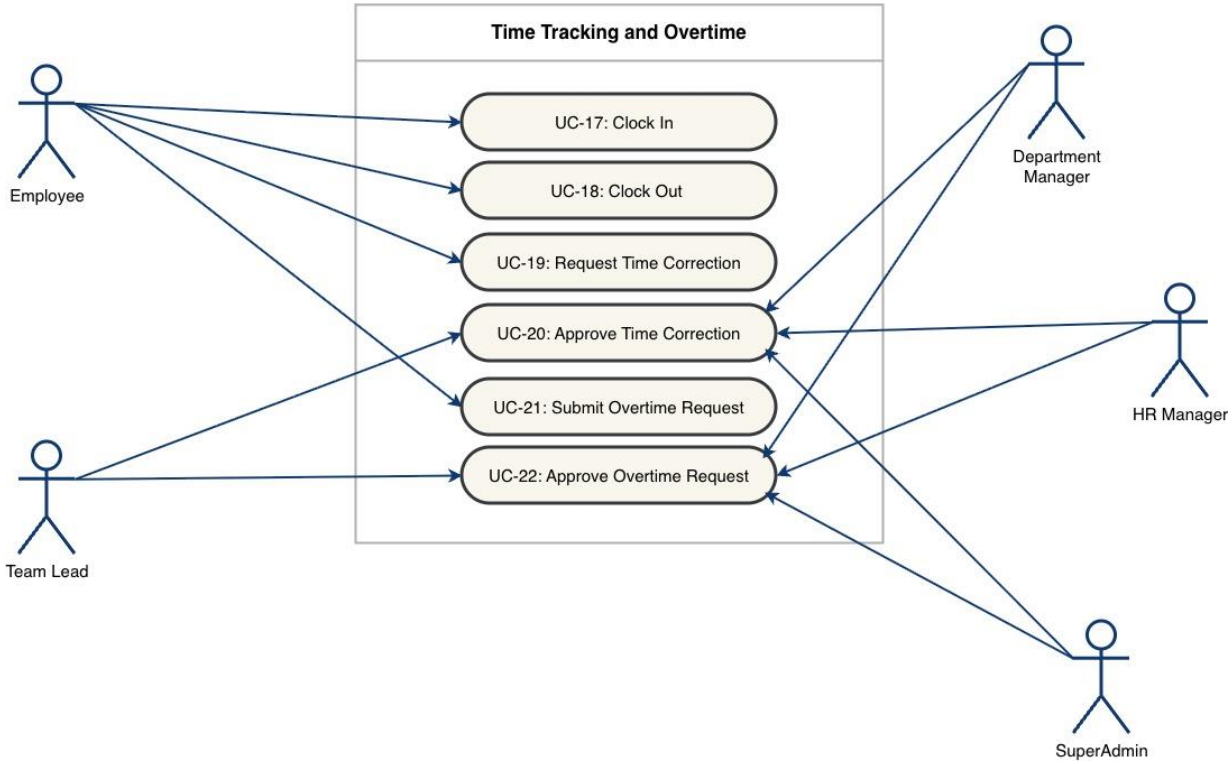

Team Lead Use Cases
Team Lead
UC-10: View Employee Directory
UC-11: Submit Leave Request
UC-12: Recommend Leave Request
UC-15: View Leave Balance
UC-17: Clock In
UC-18: Clock Out
UC-19: Request Time Correction
UC-20: Approve Time Correction
UC-21: Submit Overtime Request
UC-22: Approve Overtime Request
UC-29: Create Task
UC-30: Update Task Status
UC-31: Comment on Task
UC-35: Book a Desk
UC-42: View and Read Announcements
UC-45: Complete Onboarding Item
UC-48: Submit Manager Review
UC-49: Set Performance Goals
UC-52: View and Manage Notifications
UC-53: View Calendar

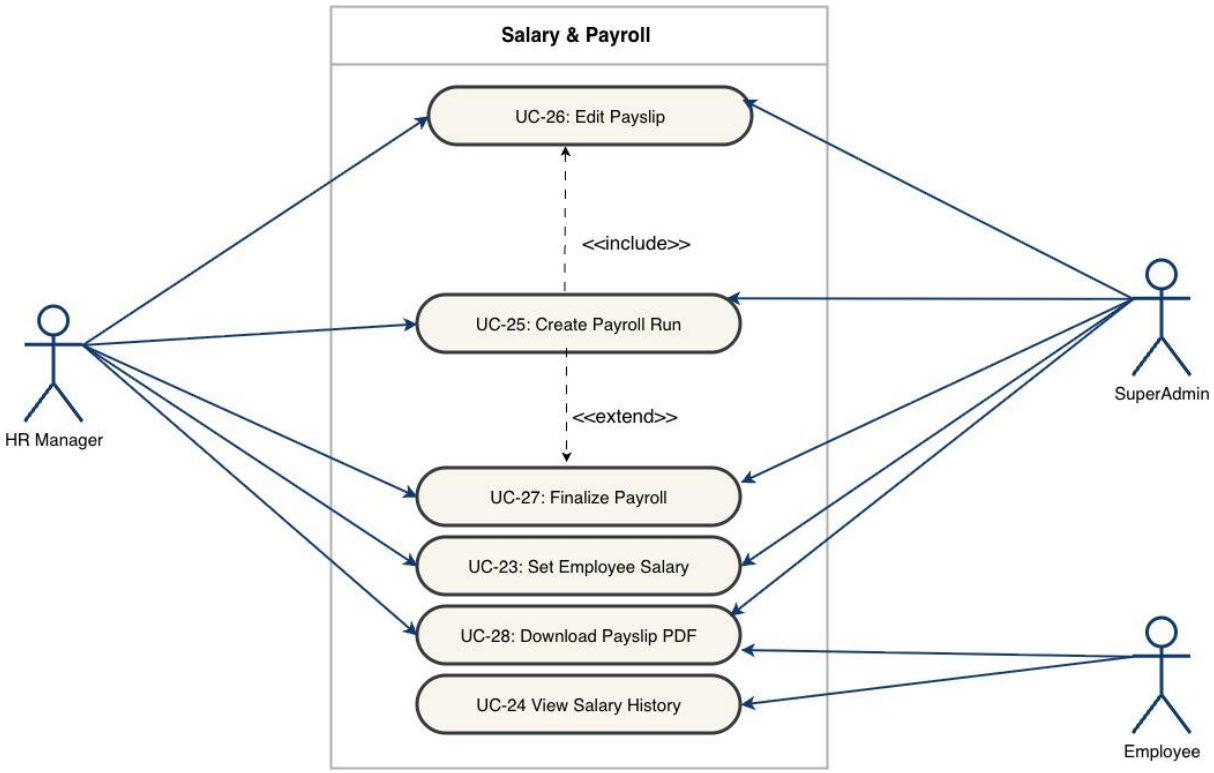

Department Manager Use Case Diagram
Department Manager
UC-10: View Employee Directory
UC-11: Submit Leave Request
UC-12: Recommend Leave Request
UC-15: View Leave Balance
UC-17: Clock In
UC-18: Clock Out
UC-19: Request Time Correction
UC-20: Approve Time Correction
UC-21: Submit Overtime Request
UC-22: Approve Overtime Request
UC-29: Create Task
UC-30: Update Task Status
UC-31: Comment on Task
UC-35: Book a Desk
UC-42: View and Read Announcements
UC-45: Complete Onboarding Item
UC-48: Submit Manager Review
UC-49: Set Performance Goals
UC-52: View and Manage Notifications
UC-53: View Calendar
UC-48: View Department Hierarchy

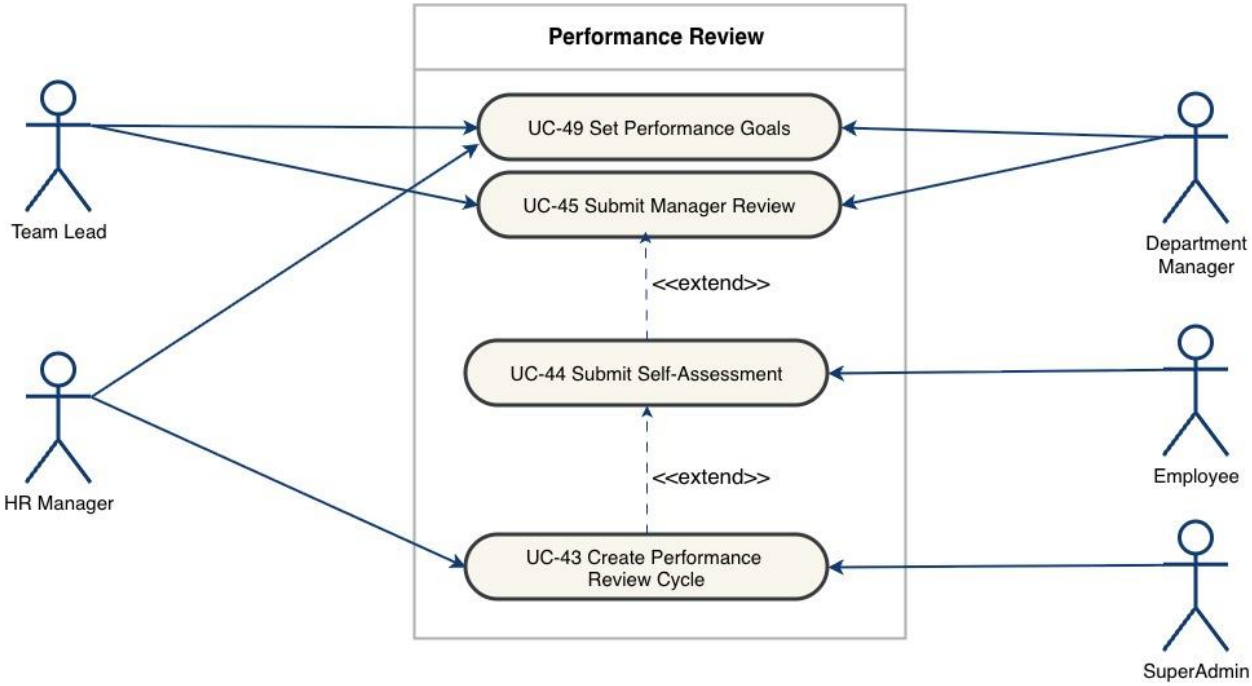

HR Manager Use Case Diagram
HR Manager
UC-05: Create Employee
UC-07: Update Employee
UC-08: Deactivate Employee
UC-09: Transfer Employee
UC-10: View Employee Directory
UC-13: Approve Leave Request
UC-15: View Leave Balance
UC-16: Configure Leave Types
UC-20: Approve Time Correction
UC-22: Approve Overtime Request
UC-23: Set Employee Salary
UC-25: Create Payroll Run
UC-26: Edit Payslip
UC-27: Finalize Payroll
UC-28: Download Payslip PDF
UC-29: Create Task
UC-30: Update Task Status
UC-31: Comment on Task
UC-32: Upload Document
UC-33: Download Document
UC-35: Book a Desk
UC-36: Cancel Desk Booking
UC-37: Manage Offices and Desks
UC-38: Create Announcement
UC-40: Create Onboarding Template
UC-41: Assign Onboarding Checklist
UC-43: Create Performance Review Cycle
UC-46: Create Department
UC-47: Update Department / Manage Head
UC-48: View Department Hierarchy
UC-50: Manage Holidays
UC-52: View and Manage Notifications
UC-53: View Calendar
UC-54: View Audit Logs

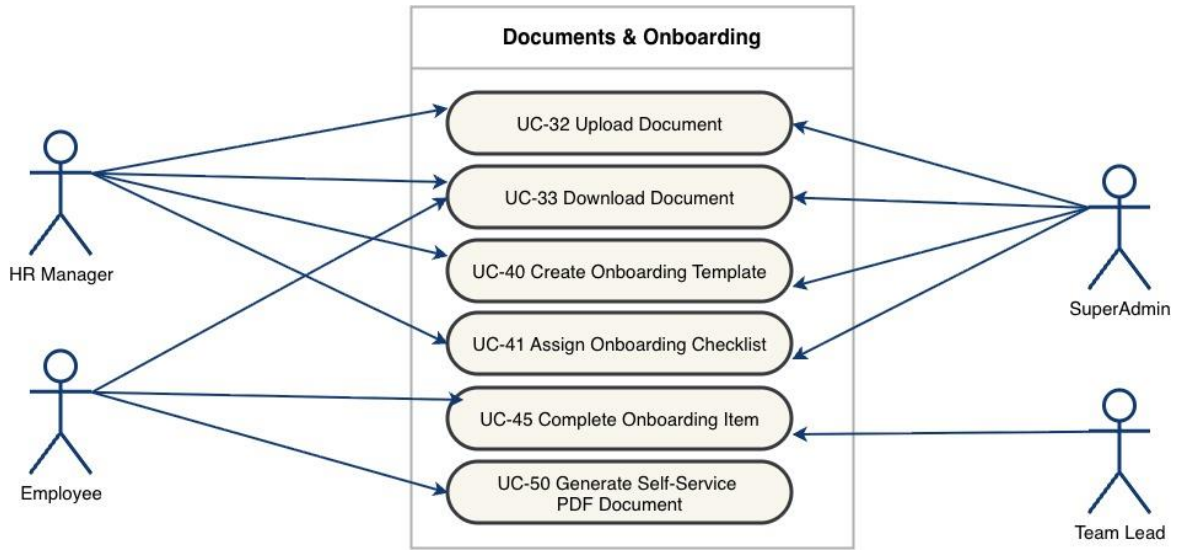

Super Admin Use Cases
SuperAdmin
UC-05: Create Employee
UC-07: Update Employee
UC-08: Deactivate Employee
UC-09: Transfer Employee
UC-10: View Employee Directory
UC-13: Approve Leave Request
UC-16: Configure Leave Types
UC-20: Approve Time Correction
UC-22: Approve Overtime Request
UC-23: Set Employee Salary
UC-25: Create Payroll Run
UC-26: Edit Payslip
UC-27: Finalize Payroll
UC-28: Download Payslip PDF
UC-29: Create Task
UC-30: Update Task Status
UC-31: Comment on Task
UC-32: Upload Document
UC-33: Download Document
UC-35: Book a Desk
UC-36: Cancel Desk Booking
UC-37: Manage Offices and Desks
UC-38: Create Announcement
UC-40: Create Onboarding Template
UC-41: Assign Onboarding Checklist
UC-43: Create Performance Review Cycle
UC-46: Create Department
UC-47: Update Department / Manage Head
UC-49: Manage User Accounts
UC-50: Manage Holidays
UC-52: View and Manage Notifications
UC-53: View Calendar
UC-54: View Audit Logs
UC-55: Manage User Accounts / Roles / Activation

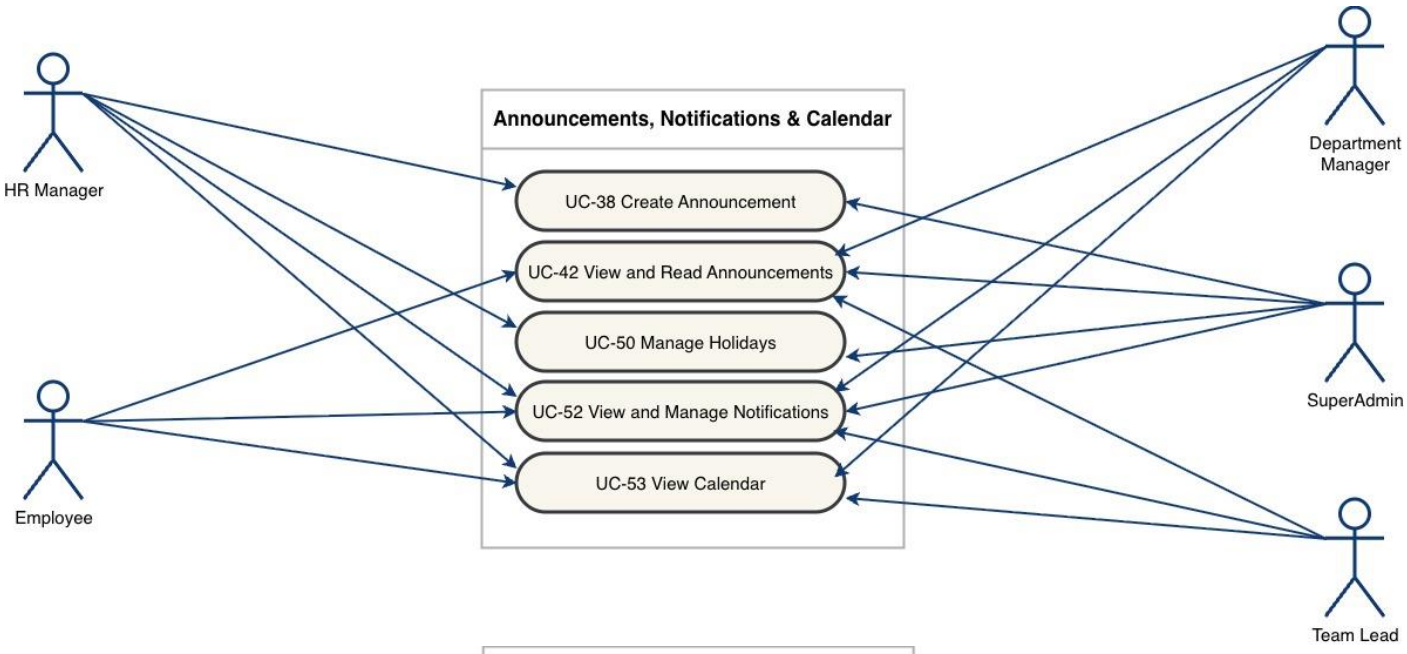

Leave Management
UC-14: Cancel Leave Request
<<extend>>
UC-11: Submit Leave Request
UC-12: Recommend Leave Request
UC-13: Approve Leave Request
UC-15: View Leave Balance
UC-16: Configure Leave Types
Employee
Team Lead
Department Manager
HR Manager
SuperAdmin

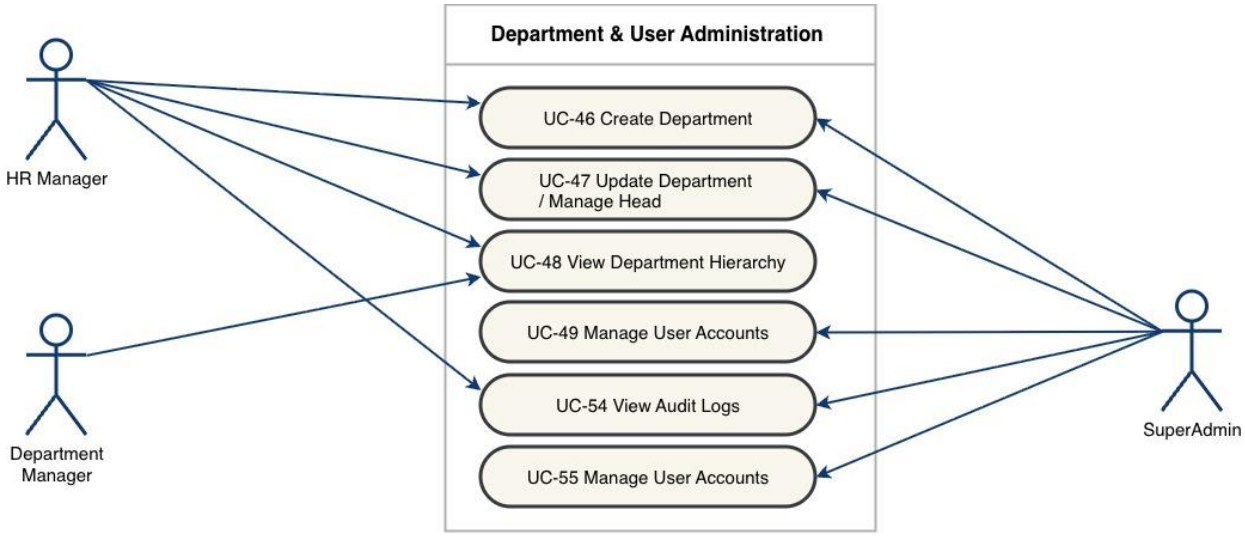

Time Tracking and Overtime
UC-17: Clock In
UC-18: Clock Out
UC-19: Request Time Correction
UC-20: Approve Time Correction
UC-21: Submit Overtime Request
UC-22: Approve Overtime Request
Employee
Team Lead
Department Manager
HR Manager
SuperAdmin


Salary & Payroll
UC-26: Edit Payslip
<<include>>
UC-25: Create Payroll Run
<<extend>>
UC-27: Finalize Payroll
UC-23: Set Employee Salary
UC-28: Download Payslip PDF
UC-24 View Salary History
HR Manager
SuperAdmin
Employee


Performance Review
UC-49 Set Performance Goals
UC-45 Submit Manager Review
<<extend>>
UC-44 Submit Self-Assessment
<<extend>>
UC-43 Create Performance Review Cycle
Team Lead
HR Manager
Department Manager
Employee
SuperAdmin


Documents & Onboarding
UC-32 Upload Document
UC-33 Download Document
UC-40 Create Onboarding Template
UC-41 Assign Onboarding Checklist
UC-45 Complete Onboarding Item
UC-50 Generate Self-Service PDF Document
HR Manager
Employee
SuperAdmin
Team Lead


Announcements, Notifications & Calendar
UC-38 Create Announcement
UC-42 View and Read Announcements
UC-50 Manage Holidays
UC-52 View and Manage Notifications
UC-53 View Calendar
HR Manager
Employee
Department Manager
SuperAdmin
Team Lead


Department & User Administration
UC-46 Create Department
UC-47 Update Department / Manage Head
UC-48 View Department Hierarchy
UC-49 Manage User Accounts
UC-54 View Audit Logs
UC-55 Manage User Accounts
HR Manager
Department Manager
SuperAdmin